Visvesvaraya National Institute of Technology, Nagpur
Department of Mechanical Engineering

A Physics-Informed Neural-Network Digital Twin of a Four-Mecanum-Wheel Omnidirectional Platform

High-fidelity-simulation accounting for slip and spin with LuGre-Hertzian friction model, adaptive sliding-mode control, and data generation with holistic excitation profiles for **forward-dynamics learning** and **inverse friction force identification**.

Vishvesh Koranne · Research Scholar
Guide: Prof. Shital S. Chiddarwar
RPC Members: Prof. Trushar B Gohil, Prof. Praveen Kumar

4 Mecnaum-Wheeled Omnidirectional Mobile Robot · slip–spin digital twin
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Outline

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CHAPTER 1 · MOTIVATION

Mecanum robot market & applications

Market growth, key sectors, and the deployment barriers that gate real-world use.

Mecanum-wheeled robot market overview

Rapid, autonomy-concentrated growth — 2024 baselines against 2030–2035 targets

Market segment	2024	Target	CAGR
Mecanum wheel systems	$1.17 B	$2.71 B (2033)	9.6%
Total mobile robots	~$5 B	$14 B (2030)	19.0%
AGV–AMR (combined)	$5.66 B	$20 B (2035)	12.1%
AMR (sub-segment)	$2.77 B	$10.36 B (2034)	15.8%

Key drivers: AGV/AMR proliferation, warehouse automation, Industry 4.0, e-commerce and labour shortages, AI navigation.

WHY THIS MATTERS FOR CONTROL

The Mecanum-wheel market grows at **9.6% CAGR to $2.71 B by 2033**; the broader mobile-robot market expands **~2× faster (19%)**. Growth is concentrated in **autonomy and control**, not the wheel itself.

At this scale, control-fidelity problems — slip, energy loss, odometry drift — become **economically significant**, not academic curiosities.

Application sectors & recurring barriers

DEPLOYMENT SECTORS

Warehouse & intralogistics
Transport, picking, sorting

Manufacturing & assembly
Feeding, material handling

Hospital & clinical logistics
Medication, sample delivery

Cleaning & inspection
Maintenance, surveillance

Assistive mobility
Wheelchairs, personal transport

FOUR RECURRING BARRIERS — BURKACKI ET AL. (2025)

Limited stability on uneven terrain
Maneuverability degrades on inclined and uneven surfaces where roller–ground contact becomes unpredictable.

High energy consumption
Wheel slippage causes measurable energy loss (6.18% documented), cutting battery-supported range proportionally.

Complex control requirements
Kinematic models omit nonlinear, load- and surface-dependent friction that pure controllers cannot reject.

Elevated cost
Precision requires external metrology (optical tracking, mechanical guides) rather than onboard control fidelity.

The barriers split cleanly: **control-related** (slip, odometry drift, disturbance rejection) versus **non-control** (cost, battery, integration, standards). **This work targets the control-related half.**

[1] M. Burkacki et al., "Systematic review of Mecanum and Omni-wheel technologies for motor impairments," *Applied Sciences* 15(9):4773, 2025.

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CHAPTER TWO · MOTIVATION

The control-fidelity gap

Physics-level obstacles the ideal-rolling model omits — and the failures they cause.

Physics-level obstacles in real-world deployment

The ideal-rolling kinematic model omits these — each has caused an independently reported failure

ENERGY LOSS FROM WHEEL SLIPPAGE — ~6.18%

A real industrial haul (5 tonnes of zinc ingots, 5 km) lost **~6.18%** of energy to slippage, cutting battery range proportionally. A kinematic controller cannot account for it. [Bayar et al., 2025]

ORIENTATION DEVIATIONS IN TRACKING

Roller–surface contact produces significant orientation deviations during tracking, needing a ramp-shaped reference intervention beyond baseline tracking to suppress. [Ortiz Hernández & Rosas Almeida, 2024]

SURFACE- & VELOCITY-DEPENDENT ODOMETRY DRIFT

Odometric drift varies with floor surface and speed — the same controller yields different positioning error across floors and speeds, undermining pose repeatability. [Shahidi et al., 2021]

COUPLED ACCURACY / ENERGY DEGRADATION

Restricted force/torque authority causes transient overshoot that simultaneously lowers tracking and raises energy use, reducing autonomy. [Ramírez-Neria et al., 2025]

ROOT CAUSE — UNDERMODELED FRICTION

Mecanum friction is comparatively undermodeled. An experimentally validated **orthotropic-friction** model shows contact behaviour depends on **both velocity and the payload-shifted centre of mass** [Manzl et al., 2024]. These effects are nonlinear, load- and surface-dependent — kinematic controllers cannot reject them.

Sector-specific control barriers

Sector	Control barrier	Evidence / impact
Heavy industry	Slip-driven energy loss shrinks range	5-t zinc haul: 6.18% loss — Bayar et al. 2025 [2]
Automotive intralogistics	Baseline controllers lack robustness	Adaptive SMC + ESO — Bui et al. 2025 [10]
Warehouse logistics	Odometry drift breaks repeatability	Surface/velocity error — Shahidi et al. 2021 [4]
Aerospace assembly	0.01 mm tolerance vs. low accuracy	Two-stage: wheels + laser — Zhao et al. 2026 [11]
Hospitals	6 risk dimensions, preliminary testing	Hygiene, interaction — Bernhard et al. 2024 [12]

BENCHMARK — KUKA OMNIMOVE

±5	±3	±1
mm base drive accuracy	mm contactless (stated)	mm with optical / mech. aids only

Key insight: deployed precision is bought by **external referencing**, not onboard control fidelity. Mecanum slip / odometry drift forces this dependence — the problem this work addresses.

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CHAPTER THREE · MOTIVATION

Related work & limitations

Higher-fidelity control closes measurable gaps — yet none recovers friction online.

Higher-fidelity model-based control: measurable gains

Gap addressed	Approach	Reported gain	Reference
Model fidelity / tracking error	Complete Lagrangian + linearized motor	17% lower DTW error	Li & Li 2025 [7]
Disturbance rejection (load, noise)	IMC + disturbance observer	97.6% / 98.3% faster settling	Li & Li 2025 [7]
Inclined surfaces (slip, gravity)	ADRC	49–60% better tracking	Ortiz Hernández 2025 [21]
Unknown CoG offset + slope	MRAC, Lyapunov adaptation	Real-time comp.; stability proof	Chaichumporn 2025 [22]
Disturbance + model uncertainty	AVPSMO feedforward in MPC	Lyapunov-proven suppression	ISA Trans. 2024 [8]

All figures are the cited authors' own results against their own baselines.

THE COMMON THREAD

Improving model fidelity and adding explicit disturbance estimation **measurably** improves tracking and robustness. The residual physics — **slip, friction, load** — must be modeled or estimated, not ignored.

THE REMAINING GAP

Yet **none recovers per-wheel friction parameters online**, and none is standard deployed practice. This is the gap the PINN + Mamba digital twin targets.

Deployed control vs. the research frontier

WHAT INDUSTRY SHIPS

- **Standard architecture:** kinematic inverse-Mecanum + per-wheel PID beneath SLAM / navigation.
- **Accuracy secured by:** mechanical precision + external referencing (optical tracking, guides).
- **KUKA omniMove:** base drive ±5 mm; contactless stated ±3 mm; ±1 mm only with optical / mechanical aids.

WHAT THE LITERATURE PROPOSES

- Adaptive integral terminal sliding mode [Sun et al., 2021].
- Anti-peaking linear ADRC [Ramírez-Neria et al., 2025].
- Sliding-mode-observer MPC [ISA Trans., 2024]; adaptive SMC + ESO [Bui et al., 2025].
- **Repeatedly show:** kinematic controllers are insufficient under real disturbance — yet none is standard practice.

THE OPENING

Close the **fidelity / identifiability gap** between deployed kinematic control and the slip/friction reality the frontier documents. **No prior work recovers per-wheel friction (μ, χ) online for Mecanum.**

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CHAPTER FOUR · MOTIVATION

PINN-based solution & its relevance

Physics-informed learning for friction and dynamics — and why interpretability enables deployment.

Physics-informed learning for friction & dynamics

LUGRE FRICTION-PARAMETER ID

A PINN fuses the LuGre dynamic friction model with learnable neural components to identify LuGre parameters from small, noisy datasets at higher fidelity than simplified models. [Ozmen et al., 2026, ACC]

JOINT DYNAMICS + RUNGE–KUTTA CELLS

A PINN encoding joint dynamics inside a recurrent net with custom RK cells outperforms conventional PINNs and grey-box SSID, matching ground-truth parameters on UR3e hardware. [Yang et al., 2023, RA-L]

ENHANCED PINNS — MOTOR & FORCE COUPLING

+31–37% modeling-accuracy gain and **−40–52%** tracking-error reduction over prior SOTA. [FITEE, 2025]

NONSMOOTH MULTI-BODY PINNS (STICK–SLIP)

PINNs embedding nonsmooth multi-body formulations capture stick–slip and separation–reattachment without the vanishingly small time steps of conventional time-stepping. [Li et al., 2024, Nonlinear Dyn.]

KAN FOR STATIC FRICTION

Kolmogorov–Arnold-network methods recover interpretable closed-form static-friction laws (**$R^2 > 0.95$**) with no predefined functional form — directly serving the interpretability requirement for certification. [Wang et al., 2025]

CRITICAL CAVEAT

Rigid-body-prior networks alone are insufficient: standard Deep Lagrangian Networks **fail** to capture nonlinear friction and flexibility, motivating augmentations (DeLaN-FFNN). This justifies pairing a physics prior **with** a learnable friction term. [Li et al., 2025]

Safety certifiability & the gap this work addresses

WHY INTERPRETABILITY MATTERS FOR DEPLOYMENT

ISO 3691-4:2023 mandates **PL d (SIL 2)** for driverless industrial trucks in shared workspaces. Black-box neural controllers are **confined to the lowest safety-integrity levels** — outputs cannot be exhaustively bounded or formally verified [Zhang & Li, 2020; Tambon et al., 2022]. A physics-informed model with known friction structure (LuGre) and interpretable per-wheel parameters **enables the verification evidence black boxes cannot supply.**

TWO OBSERVATIONS DEFINING THE OPENING

(1) The strongest PINN friction-ID and forward-dynamics evidence comes from **fixed-base manipulators**, not Mecanum platforms — transfer to per-wheel (μ, χ) recovery is established only by analogy.

(2) To our knowledge, **no prior work applies a selective state-space (Mamba) backbone** to Mecanum or mobile-robot forward-dynamics learning.

THIS PAPER'S CONTRIBUTION

A physics-informed, **Mamba-backed digital twin** that:

- Learns forward dynamics at high fidelity for a KUKA youBot Mecanum platform.
- Recovers interpretable per-wheel friction parameters (μ, χ) with physical bounds.
- Targets the **slip / friction / CoG chokepoint** that constrains Mecanum control fidelity.
- Bridges deployed kinematic control and the disturbance / slip / friction reality.

References — motivation & related work

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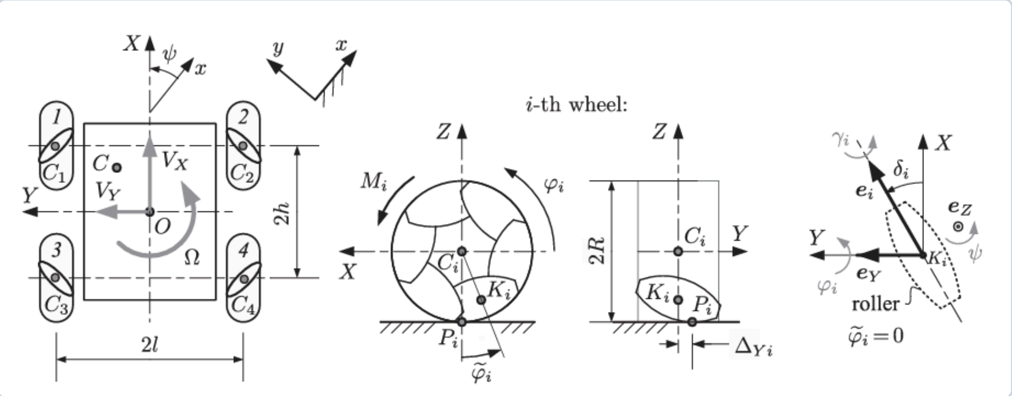
The youBot four-Mecanum-wheel platform

Adamov & Saipulaev (RJND 2020) · "Influence of dissipative forces & Mecanum-wheel design"



Plant model — Adamov & Saipulaev (2020), *Russian J. Nonlinear Dynamics* 16(2):291–307

An **omnidirectional base** with four pairwise-coaxial Mecanum wheels: independent $(V_x, V_y, \dot\psi)$ motion with no steering. Body frame $OXYZ$ at the geometric centre; heading ψ; CoM offset (a_X, a_Y) from O.

39-D SIMULATION STATE

- **1–3** body vel. $V_x, V_y, \dot\psi$
- **4** heading ψ
- **5–8** wheel angles $\theta_{1..4}$
- **9–12** wheel rates $\omega_{1..4}$
- **13–16** **roller rates** $\dot\gamma_{1..4}$ (hidden)
- **17–19** adaptive gains $K_{x,y,\psi}$
- **20–21** world position x_o, y_o
- **22–33** LuGre bristles z_x, z_y, z_s
- **34–39** super-twist DOB observer + $\hat\delta$ — **disabled this run**

Roller rates $\dot\gamma_i$ are simulated but **unmeasurable at deployment** — the PINN's GRU encoder reconstructs them implicitly. States **34–39** (super-twist disturbance observer) exist in the model but are **switched off** for this generation iteration.

Sym	Meaning	Value
$m,\ m_s$	platform / total mass	30 / 35.6 kg
I_s	yaw inertia about O	4.42 kg·m²
J_1	wheel spin inertia	5.87e−3
R	wheel radius	0.05 m
R_d	roller axle dist.	0.0355 m
$h,\ l$	half-length / -width	0.235 / 0.15 m
a_X, a_Y	CoM offset	1.6 / −2.6 cm
δ_i	roller axle angle	$\pm 45°$
p_1, p_2	viscous (wheel / roller)	0.11 / 5.78e−3
μ, χ	friction / contact-spot	*identified*

Case-1 viscous set; μ, χ are the inverse-ID targets.

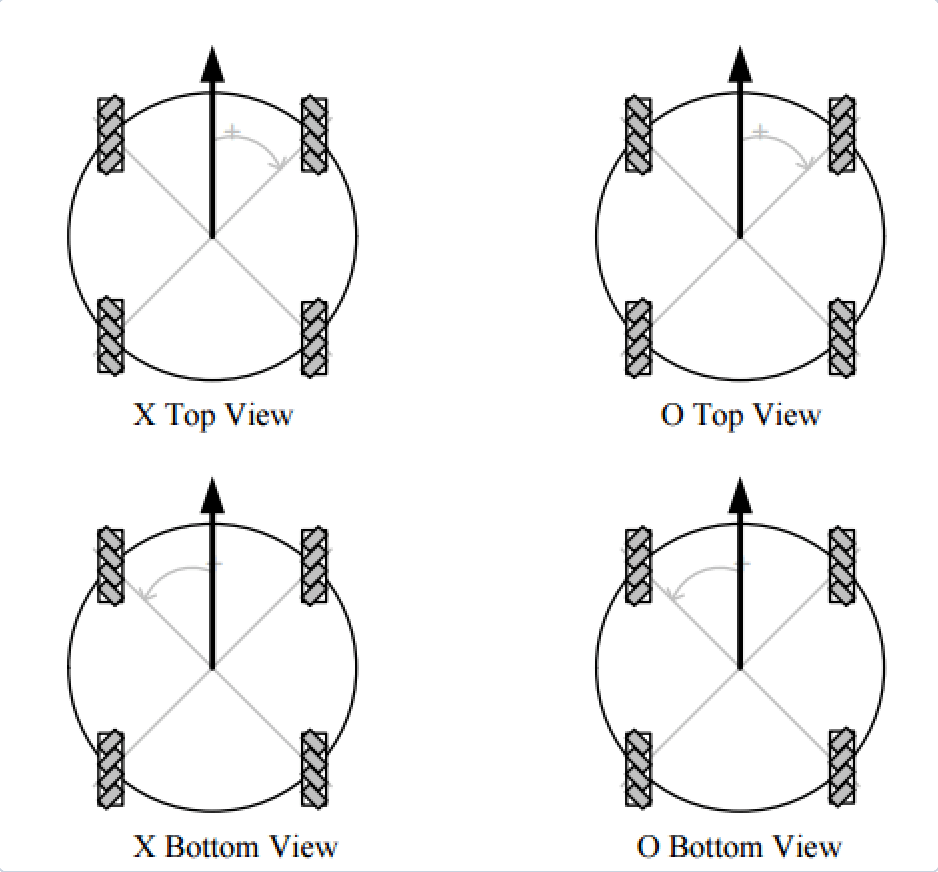
Mecanum wheels: O vs X configuration

Each wheel carries rollers skewed by $\delta_i = \pm 45°$ to the wheel plane. The **sign pattern** of δ_i around the four wheels defines the layout:

$$\begin{aligned} \text{O:}\;\; \delta &= (-,+,+,-)\,45° \\ \text{X:}\;\; \delta &= (+,-,-,+)\,45° \end{aligned}$$

This project uses the **O-configuration** $[-\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{4}, -\frac{\pi}{4}]$. O vs X swaps which diagonal roller pair points "inward," flipping the sign of the lateral- and yaw-coupling terms in the wheel↔body map.

Wheel centres $(X_i, Y_i) = (\pm h, \pm l)$; roller axle unit vector $\hat{e}_i = (\cos\delta_i, \sin\delta_i)$, rolling direction $\hat{n}_i = (-\sin\delta_i, \cos\delta_i)$.



X vs O top & bottom views — after robohub.org (drive-selection guide)

No-slip kinematics & the wheel Jacobian

Ideal rolling: each contact velocity vanishes ⇒ body twist maps to wheel rates

Per-wheel rolling constraint (contact at roller mid-section, O-config $\tan\delta_i = \pm 1$):

$$\dot{\theta}_i = \tfrac{1}{R}\Big(V_x + V_y \tan\delta_i + \dot{\psi}\,(X_i \tan\delta_i - Y_i)\Big)$$

$$\begin{bmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\\ \dot{\theta}_4\end{bmatrix} = \frac{1}{R}\begin{bmatrix}1 & -1 & -(h+l)\\ 1 & 1 & (h+l)\\ 1 & 1 & -(h+l)\\ 1 & -1 & (h+l)\end{bmatrix}\begin{bmatrix}V_x\\ V_y\\ \dot{\psi}\end{bmatrix}$$

PASSIVE ROLLER NO-SLIP RELATION

$$\dot{\varphi}_i = \frac{V_Y + X_i\dot{\psi}}{(R_d - R)\cos\delta_i}$$

The wheel nulls body-x slip (lever $p_y = \pm l$); the **passive roller spin** $\dot{\gamma}_i$ nulls body-y slip (lever $p_x = \pm h$).

INVERSE: WHEEL TORQUES → BODY WRENCH

$$F_p = \Big(\tfrac{1}{R}\sum\tau_i,\ \tfrac{1}{R}(-\tau_1+\tau_2+\tau_3-\tau_4),\ \tfrac{l+h}{R}(-\tau_1+\tau_2-\tau_3+\tau_4)\Big)$$

The full plant departs from this ideal: rollers slip and spin, and the contact point migrates — the next section.

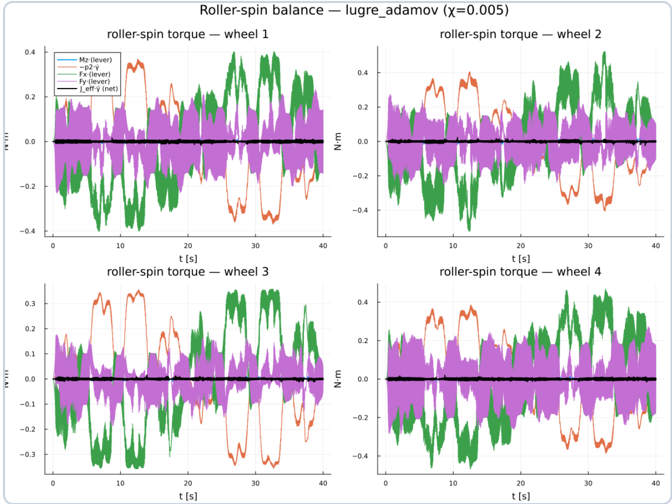
Why model roller slip and spin

Real Mecanum wheels are **discrete**: one roller touches the floor at a time, handing off as the wheel turns. The no-slip ideal erases three real effects:

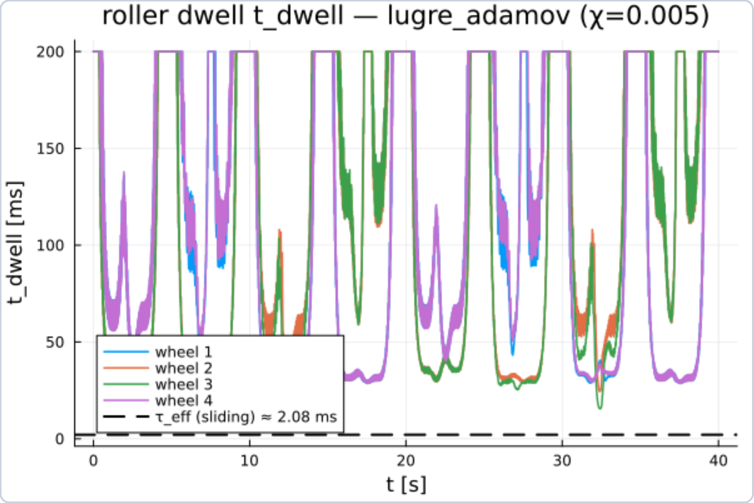
- **Slip velocities** at P_i drive the friction: $V_{PiX} = V_x - \rho_{Yi}\Omega - \dot{\varphi}_i R + \dot{\gamma}_i \sin\delta_i(R_d \cos\tilde{\varphi} - R) - \ldots$
- **Passive roller DOF** $\dot{\gamma}_i$: sustained only by contact friction vs drag p_2 (J_{roller}=10^{-6}) — **quasi-static** (dwell $\gg \tau_{\text{eff}}$, lower right).
- **Contact migration** $\Delta Y_i = R_d \tan\tilde{\varphi}_i \tan\delta_i$: a parametric modulation at the roller-passing frequency.

WHY IT MATTERS FOR THE TWIN

Wheel/roller dissipation reflects into **equivalent body damping** on $V_y, \dot{\psi}$; the PINN recovers μ, χ from body-frame signals and its GRU reconstructs the hidden $\dot{\gamma}_i$.



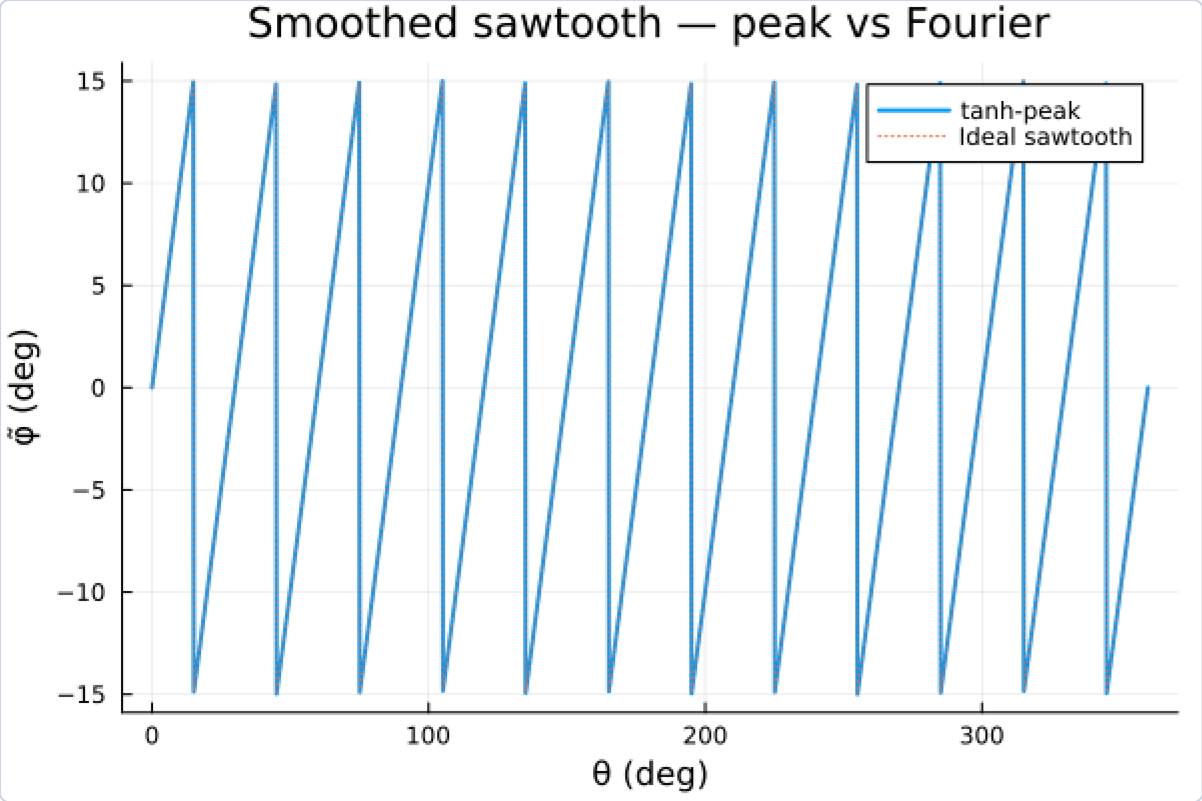
Roller-spin torque balance — M_z lever leads; net $J_{\text{eff}}\dot{\gamma} \approx 0$.



Roller dwell $t_{\text{dwell}} \gg \tau_{\text{eff}} \approx 2$ ms $\Rightarrow$ quasi-static slaving.

The roller hand-off — 12 rollers per wheel

Our model uses 12 rollers/wheel (the papers use 6) for finer numerical fidelity



The contacting-roller angle $\tilde{\varphi}_i$ ramps within one roller's sector, then jumps at hand-off:

12 rollers (ours): $\tilde{\varphi} = \big((\theta + 15^\circ) \bmod 30^\circ\big) - 15^\circ$

6 rollers (paper): $(\theta + 30^\circ) \bmod 60^\circ - 30^\circ$

- **12 vs 6:** halves the sector ($\pm 15^\circ$) and **doubles** the hand-off frequency.
- **Solver-safe smoothing:** the ideal sawtooth is replaced by a C^∞ tanh-peak $\big(\operatorname{arctan2}(K \sin 12\theta,\ K \cos 12\theta + 1)/12,\ K = 60\big)$ — no Jacobian discontinuities for the stiff implicit integrator.

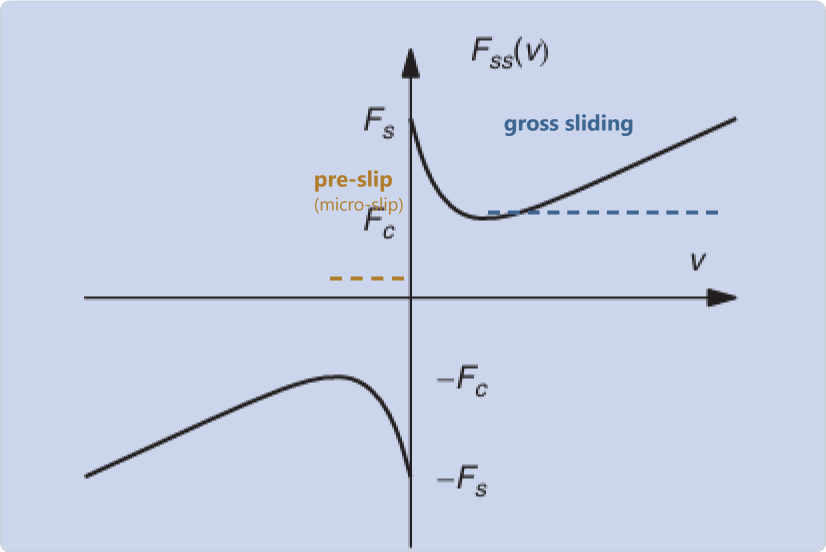
The controller drives the plant into pre-slip

Under closed-loop adaptive sliding-mode control, roller slip is held **below the Stribeck transition** — in the microslip / pre-sliding regime — over the **majority** of every trajectory. A concrete dynamic model *in that region* is therefore not optional.

WHY STATIC / COULOMB / ADAMOV FAILS HERE

- **Adamov multicomponent law** is derived under **gross sliding** (every contact element slides) — that assumption is violated in pre-slip.
- **Coulomb** is **discontinuous at zero slip**: direction undefined, magnitude jumps $f_s N \to 0$; it cannot represent stick, hysteretic pre-sliding displacement, or smooth reversals.
- The discontinuity injects **Jacobian breaks** that destabilise the stiff implicit solver.

Resolution: a LuGre **bristle** state z gives a smooth, hysteretic pre-sliding law that **recovers Adamov exactly at gross sliding** and reduces to LuGre at full stick.



LuGre friction curve — after Åström & Canudas-de-Wit (2008), *IEEE Control Systems Mag.* 28(6):101–114; pre-slip / gross-slip regions marked.

The composite LuGre–Adamov coupled model

Dynamic bristles (LuGre) × slip–spin coupling (Adamov) × Hertz–Mindlin stiffness

Bristle dynamics (linear z_x, z_y + rotational z_{rot}):

$$\dot{z}_x = V_{px} - \sigma_0 \frac{V_{\text{eff},xy}}{g_{xy}} z_x, \qquad \dot{z}_{\text{rot}} = \omega_z - \sigma_{0,r} \frac{V_{\text{eff},mz}}{g_{mz}} z_{\text{rot}}$$

Stribeck capacity (rubber $\alpha = \frac{1}{2}$): $g_{xy} = \left[f_c + (f_s - f_c)e^{-(V_{\text{eff},xy}/v_s)^{\alpha}}\right] N$

Slip–spin coupling (Mindlin annulus ramp $\mathcal{R}$):

$$V_{\text{eff},xy} = \|V_p\| + \mathcal{R}_{xy} \tfrac{8}{3\pi} |\omega_z| a, \quad V_{\text{eff},mz} = \tfrac{16}{3\pi} |\omega_z| + \mathcal{R}_{mz} \tfrac{5\|V_p\|}{a}$$

OUTPUT FORCE & SPIN TORQUE

$$F_x = \sigma_0 z_x + \sigma_1 \dot{z}_x, \qquad M_z = \sigma_{0,r} z_{\text{rot}} + \sigma_{1,r} \dot{z}_{\text{rot}}$$

IDENTIFIED PARAMETERS

$\mu \leftrightarrow f_c/f_s$ friction family · $\chi \leftrightarrow$ contact-spot scale (Hertz radius $a = (3NR_{\text{roller}}/4E^*)^{1/3}$). Stiffnesses $\sigma_0 = 8G^*a$, $\sigma_{0,r} = \frac{16}{3} G^* a^3$ precomputed per wheel.

Limits recovered: gross sliding $\rightarrow$ Adamov $F_{ss} = -f_c N\, V_p / (\|V_p\| + \frac{8}{3\pi}|\omega_z| a)$; full stick $\rightarrow$ LuGre. C^∞ throughout.

Calibration, the Mindlin ramp & the slip–spin budget

BRISTLE CONSTANTS (LOAD-NORMALISED)

	σ_0 [1/m]	σ_1 [s/m]
translation	1.64×10^3	1.6
spin	1.09×10^3	1.1

$F = -N(\sigma_0 z + \sigma_1 \dot{z} + \sigma_2 V_p)$; $M_z = -N\chi^2(\sigma_{0,s} z_s + \sigma_{1,s}\dot{z}_s + \dots)$ – spin torque scales with χ^2.

σ_1 FROM A BRISTLE RELAXATION TIME

σ_1 is fixed by an assumed **relaxation time** $\tau \approx 1$ ms via $\sigma_1 = \sigma_0\,\tau$: $1.6 = 1640\,\tau$ and $1.1 = 1090\,\tau \Rightarrow \tau \approx 0.98$–$1.01$ ms (both channels consistent).

MINDLIN SLIP-ANNULUS RAMP — COUPLING TURNS ON WITH SLIP

$$\mathcal{R} = 1 - \left(1 - \frac{\|z_{xy}\|}{\delta^*}\right)^{2/3}, \qquad \delta^* = \mu_s/\sigma_0$$

As bristle deflection grows ($\|z\| \to \delta^*$), $\mathcal{R} : 0 \to 1$: at **full stick** the cross-coupling is off (pure LuGre); approaching **gross slide** the full Adamov spin↔slip coupling switches on (C^∞).

COUPLING DIVIDES THE FRICTION BUDGET

Slip and spin share one Coulomb circle through the effective-slip denominators $D_{xy} = \|V_p\| + \frac{8}{3\pi}|\omega_z|a$, $D_{mz} = 5\|V_p\| + \frac{16}{3\pi}|\omega_z|a$:

translation-dominant $\Rightarrow F \to f_c N,\ M_z \to 0$; spin-dominant $\Rightarrow F \to 0,\ |M_z| \to \frac{3}{16} f_c N a$.

The coupling **apportions** the finite budget between slip force and spin torque.

Adaptive sliding-mode control of the platform

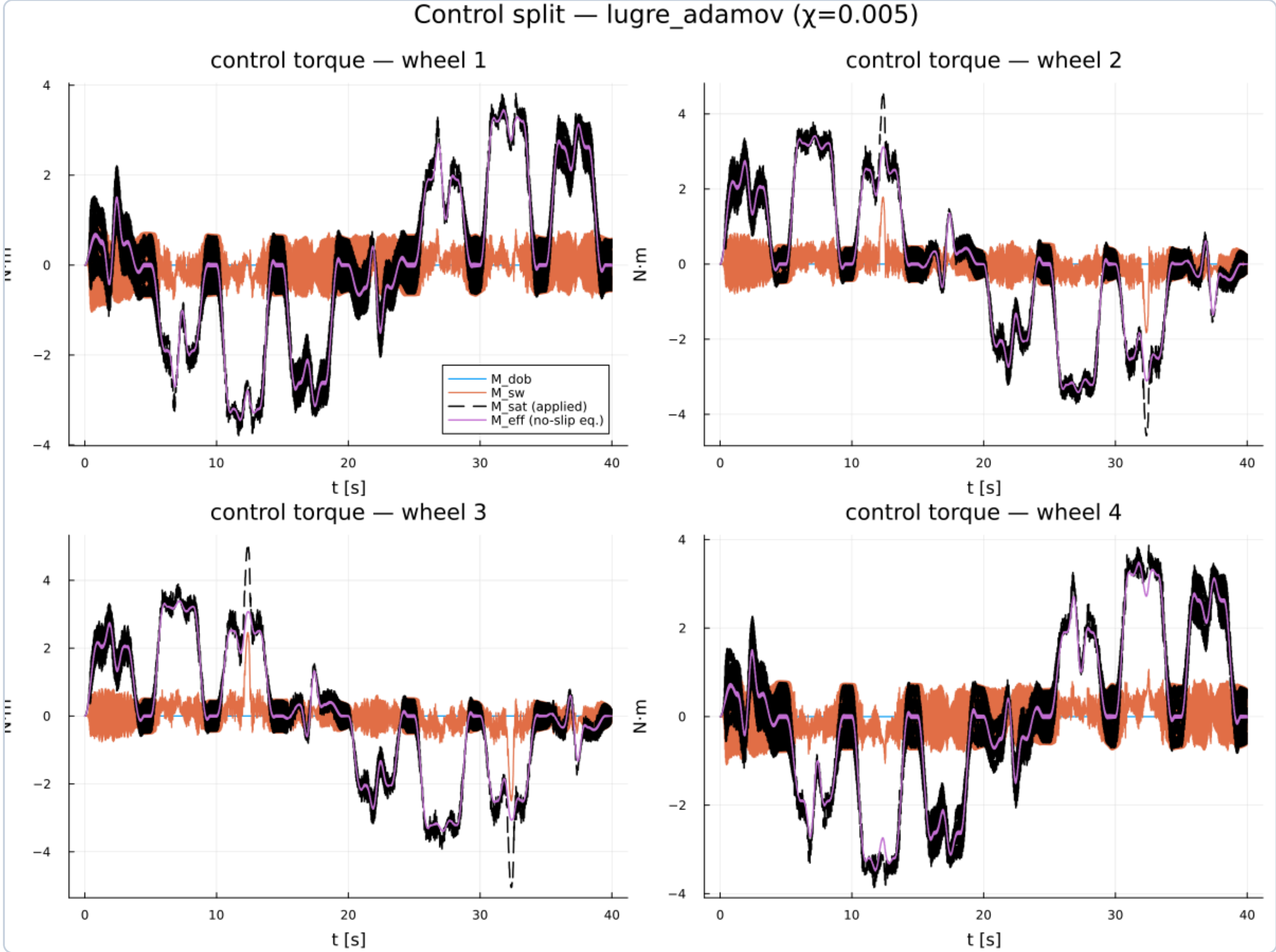
- **VelRef** (**asmc_torques_vel**): **mixed relative-degree** — degree-1 velocity surfaces on (V_x, V_y), degree-2 heading surface on ψ (6 of 7 profiles).
- **PosRef** (**asmc_torques**): degree-2 world-position tracking (ellipse).

$$s_x = V_x - V_x^d, \quad s_y = V_y - V_y^d, \quad s_\psi = \dot{e}_\psi + \lambda_\psi e_\psi$$

AUGMENTED MASS MATRIX

$$M_{\text{aug}} = \begin{bmatrix} \tilde{m} & 0 & -ma_Y \\ 0 & \tilde{m} & ma_X \\ -ma_Y & ma_X & I_{\psi,\text{aug}} \end{bmatrix}$$

$\tilde{m} = m_s + 4J_1/R^2$; equiv. control inverts M_{aug} ($\sim$13% coupling). Gain law $\dot{K}_i = \gamma_i \sigma_i b(K_i) - c_i (K_i/K_{\max,i})^3$ – leak, $K_{\max} = (60, 80, 100)$.



Per-wheel control split (octagon run): equivalent M_{eff}, switching M_{sw}, applied M_{sat}.

Uniform ultimate boundedness of both controllers

Velocity-tracking stability note (PoseTrack 2) — disturbance-observer disabled

On the manifold each surface collapses to the canonical scalar form

$$\dot{s}_i = -\tilde{K}_i \tanh(s_i/\varepsilon_i) + \tilde{d}_i, \qquad |\tilde{d}_i| \le D_i^{\mathrm{eff}}$$

With $V_i = \frac{1}{2}s_i^2 + \frac{\beta_i}{2\gamma_i}(K_i - K_i^*)^2$, the reaching condition is on the **raw** gain (the projection β_i cancels):

$$\boxed{K_i^* > \frac{D_i^{\mathrm{wr}}}{\tanh 1} \approx 1.313\, D_i^{\mathrm{wr}}}$$

$\Rightarrow$ every trajectory enters a ball $|s_i| \le \varepsilon + O(D/K_{\max})$ in finite time and stays. Cubic push-back + leak keep K_i bounded.

DEGREE-2 YAW SURFACE & GAIN ADAPTATION

$$s_\psi = \dot{e}_\psi + \lambda_\psi(e_\psi)\, e_\psi, \qquad \dot{s}_\psi = -\tilde{K}_\psi \tanh \frac{s_\psi}{\varepsilon_\psi} + \tilde{d}_\psi$$

On the manifold $\dot{e}_\psi = -\lambda_\psi e_\psi$ (**linear damping**) — a biased s_ψ no longer integrates, so a circle stays a circle (no spiral drift).

$$\dot{K}_\psi = \underbrace{\gamma_\psi \sigma_\psi b(K_\psi)}_{\text{growth}} - \underbrace{c_\psi \left(\frac{K_\psi}{K_{\max,\psi}}\right)^3}_{\text{cubic pushback}} - \underbrace{\sigma_\psi^*(K_\psi - 0.95 K_{\psi 0})\, g}_{\text{linear leak}}$$

DISTURBANCE HEADROOM (NO DOB)

axis	$K_{\max}$	adm. $D = \tanh 1 \cdot K_{\max}$
x	60	45.7 N·m
y	80	60.9 N·m
ψ	100	76.2 N·m

How the caps are set: the friction circle

Analytical limits note (AxisVel / AccelEnvelope) — what bounds the excitation

Each contact force lives on a circle $F_{\parallel}^2 + F_{\perp}^2 \le (\mu N_i)^2$:

- $F_{\parallel}$ (axle): wheel traction — **motor-supplied**.
- $F_{\perp}$ (rolling): roller-spin drag — **velocity-slaved**, consumes the circle radially.

cap	mechanism	value
$V_{x,\mathrm{cap}}$	torque / drag (soft)	4.55 m/s
$V_{y,\mathrm{crit}}$	friction circle (hard)	0.63 m/s
$\dot\psi_{\max}$	$F_{\perp}$ fills circle	3.8 rad/s
$a_x, a_y, \ddot\psi$	per-wheel at rest	2.93, 2.86, 9.6

Anisotropy $V_{x,\mathrm{cap}}/V_{y,\mathrm{crit}} \approx 7$ — a near-vertical torque wall vs a rounded friction fold. Master envelope & binding-wheel gate at right.

MASTER PER-WHEEL ENVELOPE — EQ. (27)

$$\left(A_{\parallel}^{+}\vec b\right)_i^2 + F_{\perp,i}(\vec v)^2 \le (\mu N_i)^2, \quad i = 1\ldots4$$

$\vec b = M\dot{\vec v} + C(\dot\psi)\vec v - A_n\vec F_{\perp}(\vec v)$: the velocity-slaved drag $F_{\perp}$ is radial (fixed by $\vec v$); the inertial demand sets the traction $F_{\parallel}$ (residual).

REAR-LEFT BINDING-WHEEL GATE — EQS. (30)–(32)

$$F_{\perp,3} = \kappa\,(V_y - h\dot\psi), \qquad \kappa = \frac{\sqrt{2}\,p_2}{(R-R_a)^2} = 38.9$$

$$F_{\parallel,3} = 0.354\,(b_x + b_y) - 0.918\,b_\Omega$$

$$\text{feasible} \iff F_{\parallel,3}^2 \le (\mu N_3)^2 - F_{\perp,3}^2, \quad \mu N_3 = 34.8\ \mathrm{N}$$

Derived from the **no-slip (Model N) reduction**; the lightest wheel (μN_3) binds first, collapsing the four-wheel test to a single scalar gate.

Octagon — directional sweep at low load

MOTION

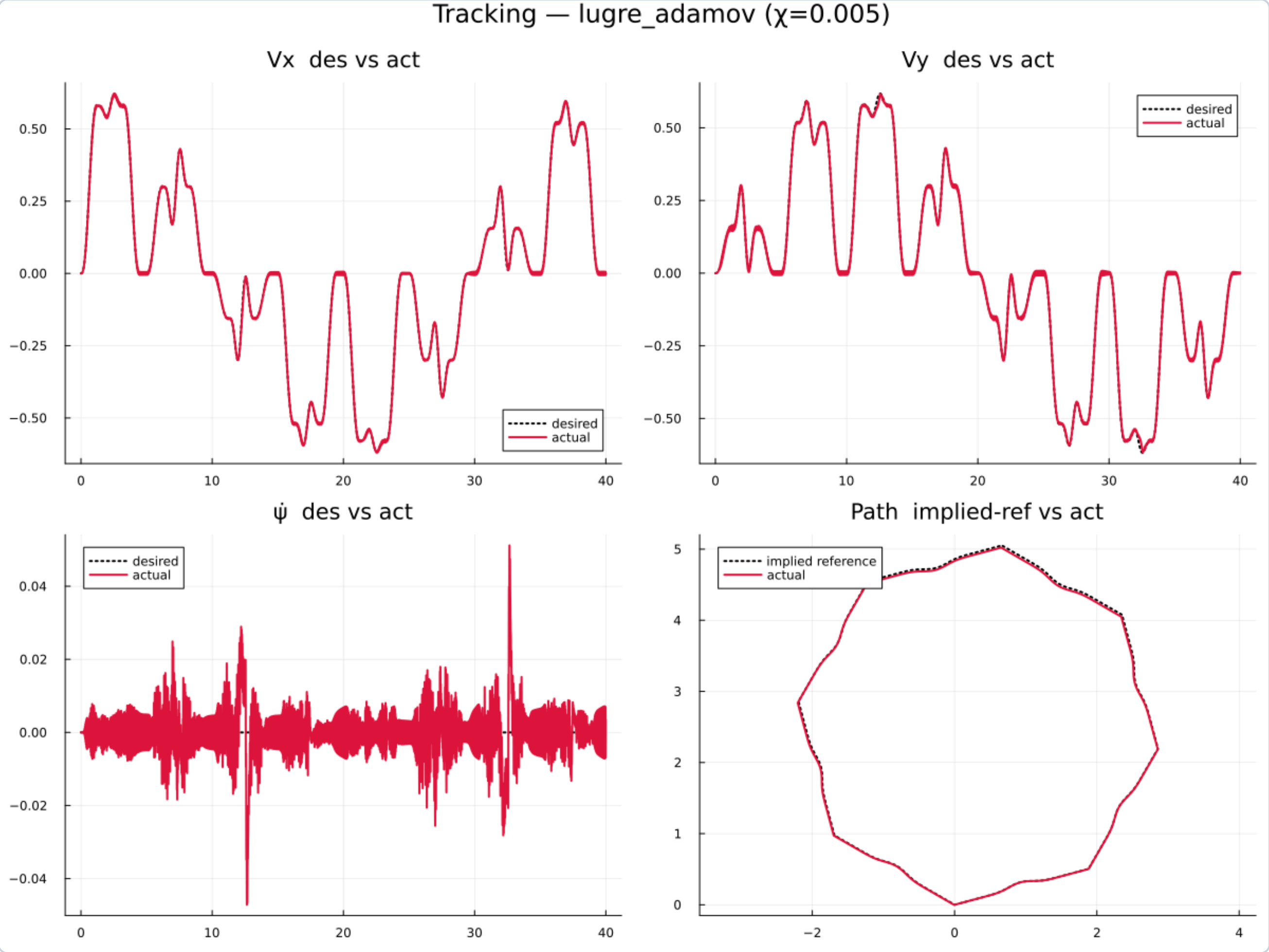
Start–cruise–stop body-velocity legs in N directions (heading held fixed, $\Omega \equiv 0$); a confined lateral wiggle rides each cruise (C^2 at the edges).

FRICTION-CIRCLE LOADING

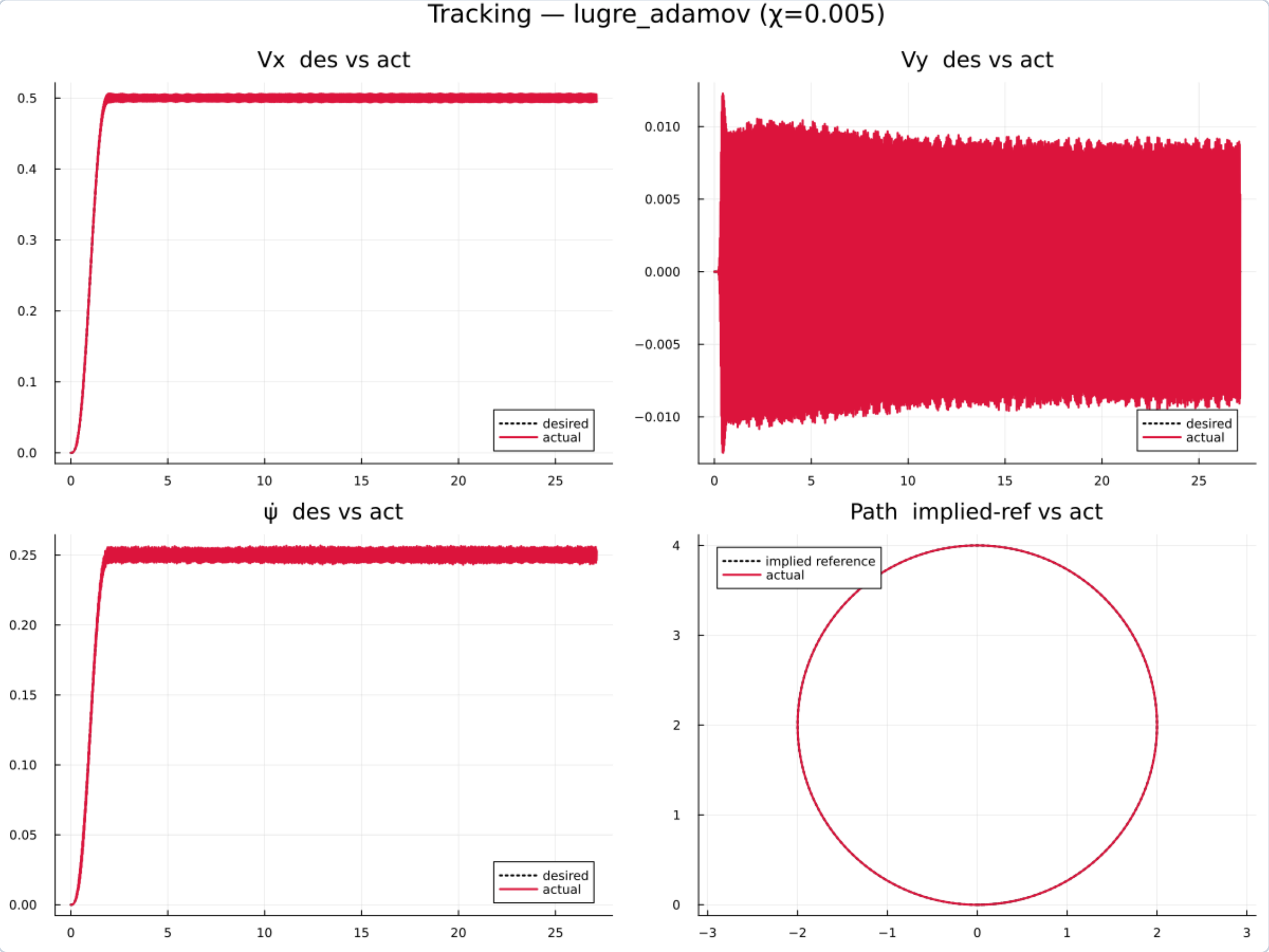
Forward legs spin no rollers $\Rightarrow F_\perp \approx 0$, circle nearly **empty** (motor-limited). The lateral wiggle injects small $F_\perp$.

SIGNIFICANCE

Azimuthal coverage of the force directions at low utilization, with controlled lateral excursions toward the V_y wall.



Long circle — sustained forward rolling



MOTION

Heading-aligned orbit: $V_x = R\dot{\psi},\ V_y = 0$, heading sweeps **with** the tangent. World velocity traces a clean circle — no drift, no position tracking.

FRICTION-CIRCLE LOADING

Forward-rolling $\Rightarrow$ steady $F_\parallel$, small $F_\perp$ (only the yaw-lever spin). Low, **quasi-static** utilization.

SIGNIFICANCE

Long stationary friction state — excites the **LuGre steady-state** (Stribeck) force cleanly; repeatable laps.

Spin-creep — the yaw-rate edge

MOTION

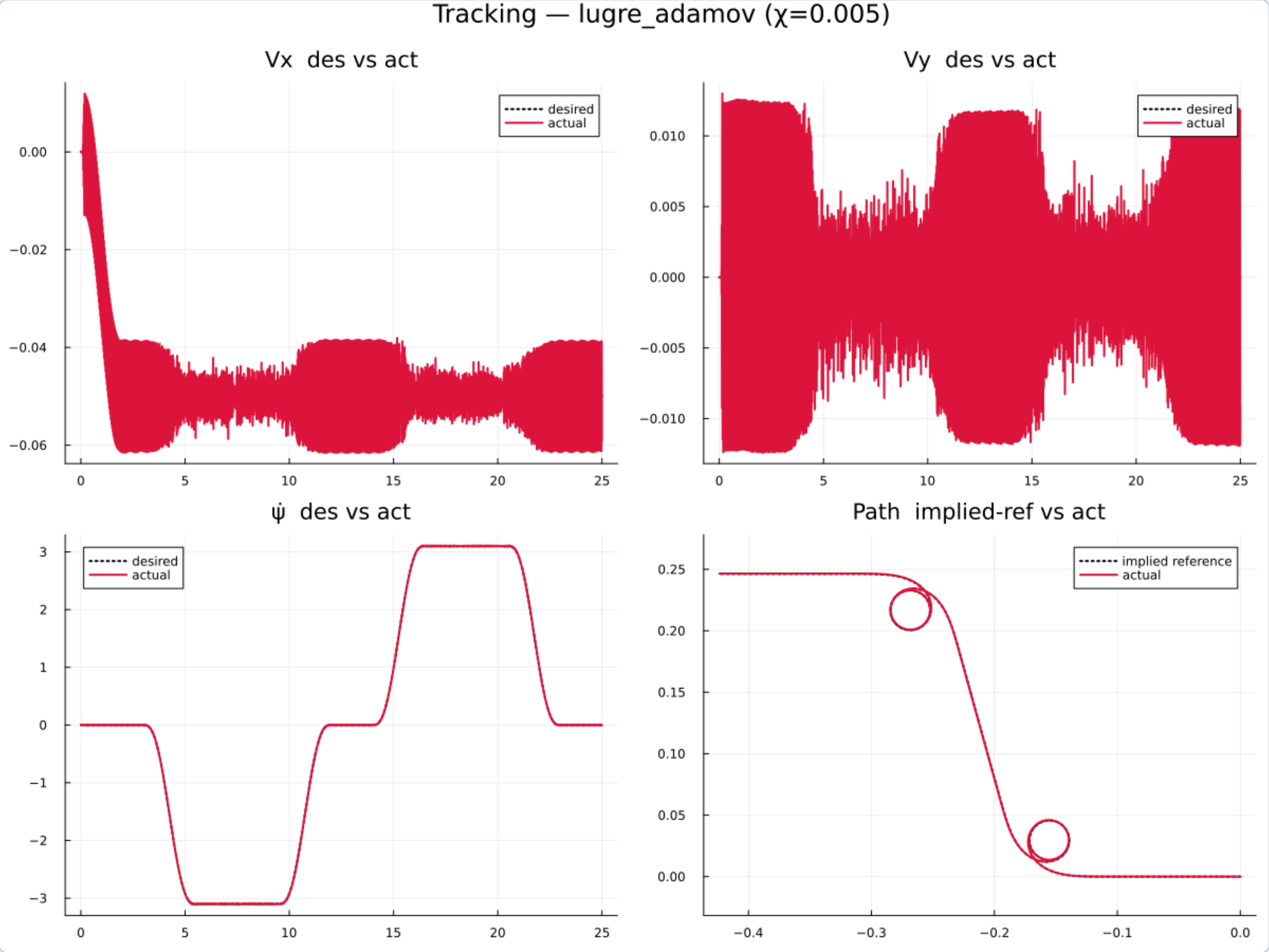
Small constant creep translation + scheduled **high yaw-rate pulses**. Low $|V|$ frees the wheel-speed budget for large $|\Omega|$.

FRICTION-CIRCLE LOADING

Spin-dominated: the spin lever h drives large roller drag $F_{\perp}$ that **fills the circle radially** toward $\dot{\psi}_{\max} \approx 3.8$ rad/s; $F_{\parallel}$ small.

SIGNIFICANCE

Probes the **rotational** friction limit; exercises the rotational bristle z_{rot} and spin torque M_z (the χ-sensitive channel).



Coupled V–Ω — rays through the interior

MOTION

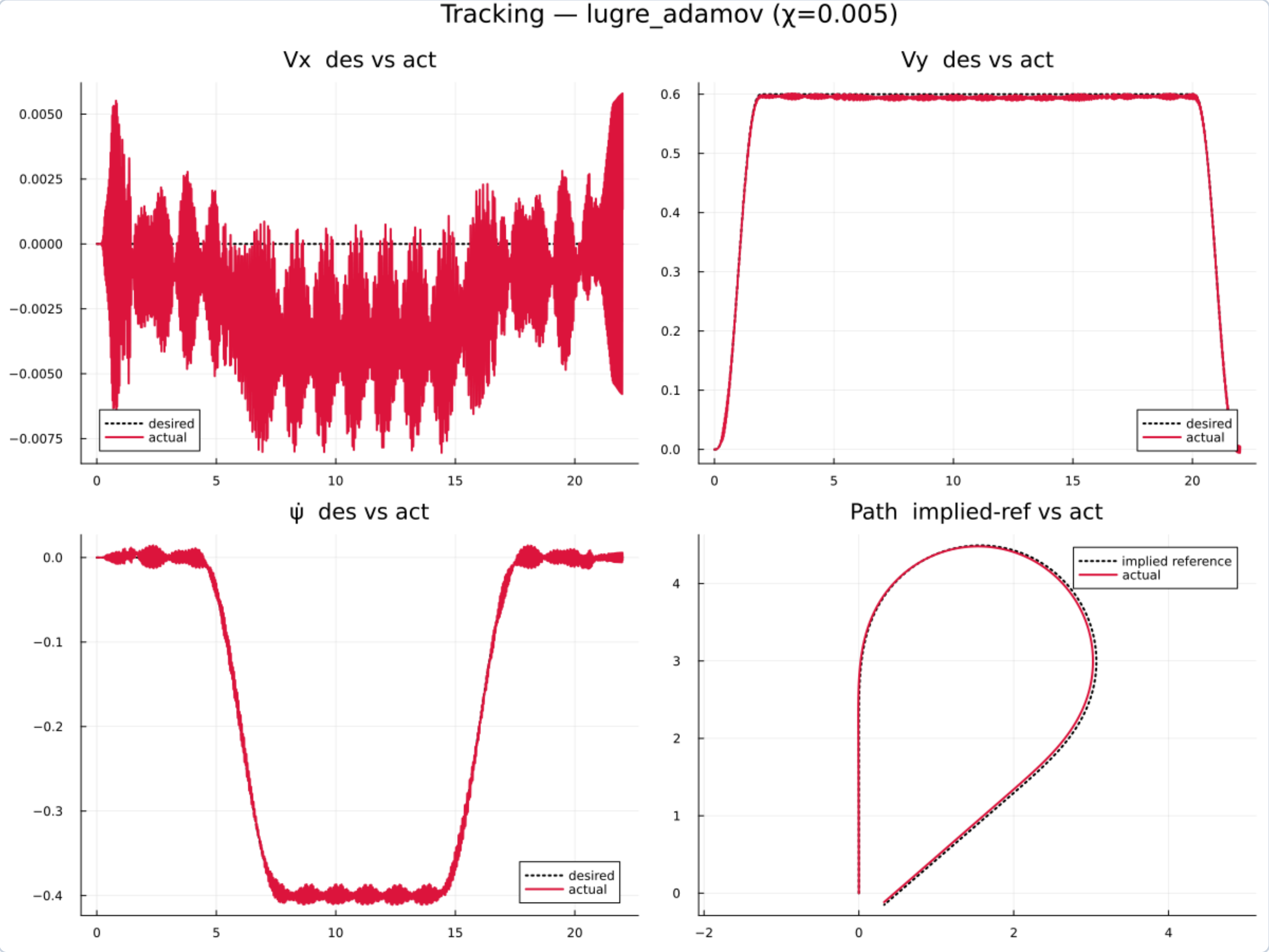
Independent $|V|$ and Ω profiles, each held or ramped (const / ramp). The state rides a chosen (V, Ω) ray through the reachable set.

FRICTION-CIRCLE LOADING

Mixed: $F_{\parallel}$ from speed/acceleration and $F_{\perp}$ from spin load the circle along an **oblique ray** — the interior coupling.

SIGNIFICANCE

Decorrelated V–Ω content; samples combined-motion states the marginal profiles miss.



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Spiral orbit — sustained near-boundary

MOTION

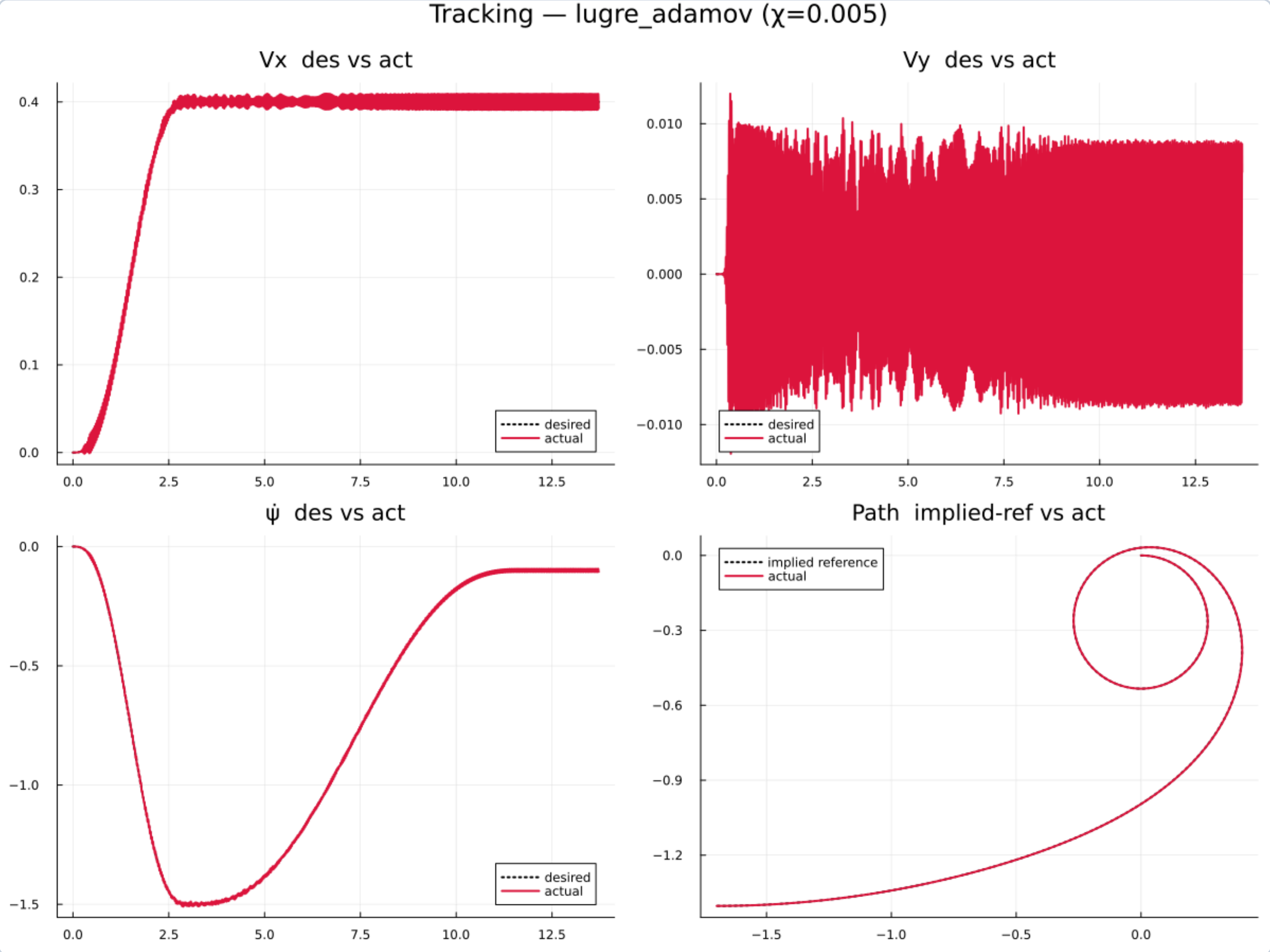
Heading-locked geometric spiral, $V = |\Omega| R(t)$. Three modes: **om_const** (sweep V), **v_const** (sweep Ω), **iso_accel** (hold $u^* = |\Omega| V$).

FRICTION-CIRCLE LOADING

Operates **near the envelope edge**; iso_accel holds centripetal utilization u^* constant just inside the cap while (V, Ω) composition sweeps.

SIGNIFICANCE

Traces the **boundary** systematically — V-, Ω-, and composition-marginals near saturation; unique in the set.



Multisine — broadband persistent excitation

MOTION

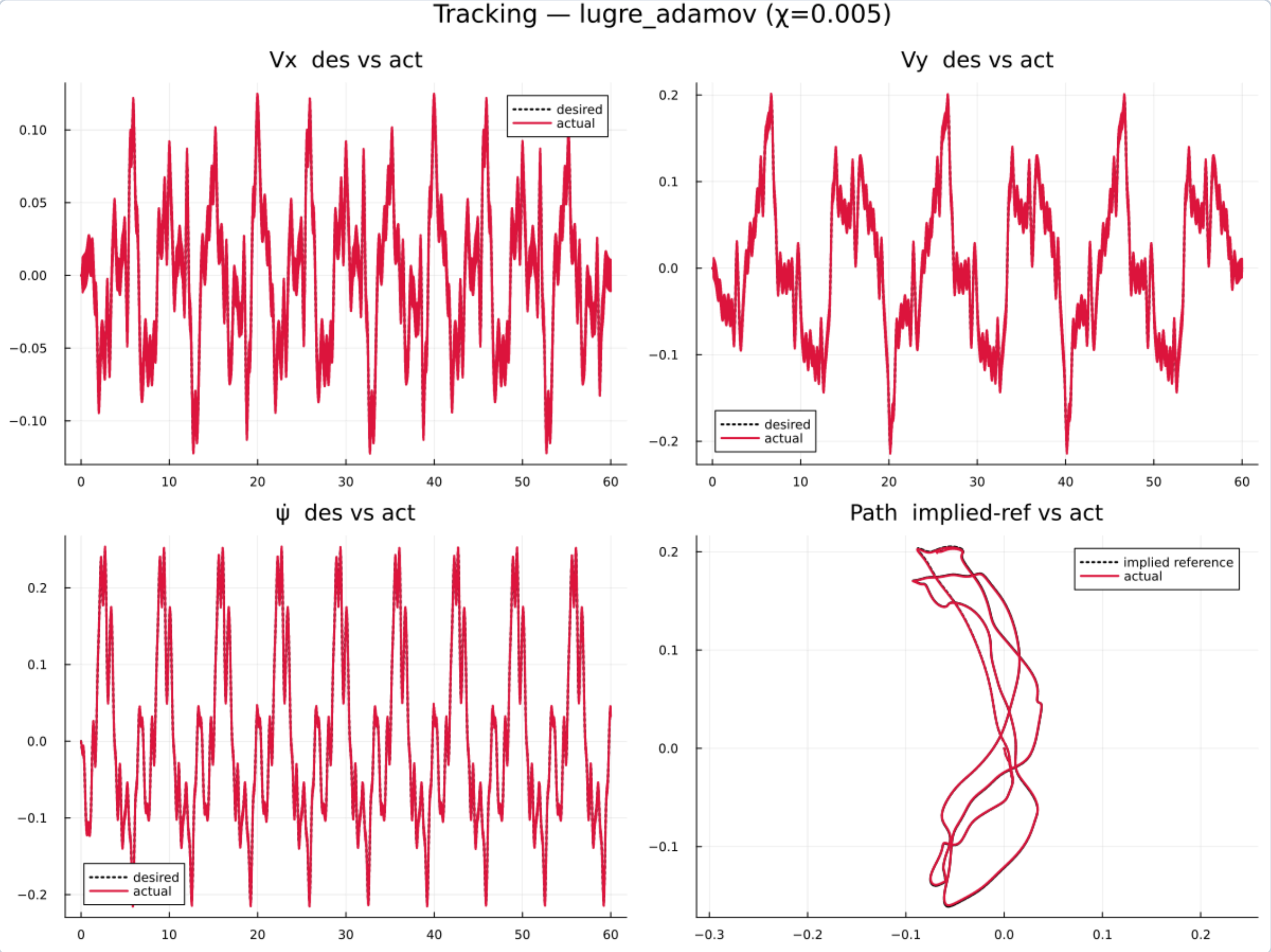
Zero-mean harmonic sums on $V_x, V_y, (\Omega)$ at integer harmonics of $1/T_{\text{fund}}$ (periodic, bounded CoM). **Zippered** disjoint combs per channel.

FRICTION-CIRCLE LOADING

Broadband transient $F_{\parallel}$ (acceleration) and $F_{\perp}$ (spin) — scattered, fluctuating load across the circle and across frequency.

SIGNIFICANCE

Persistent excitation with decorrelated joint (V_x, V_y, Ω) coverage; excites the LuGre **dynamic** (bristle) transients across the band.



Ellipse — position tracking & the lateral wall

MOTION

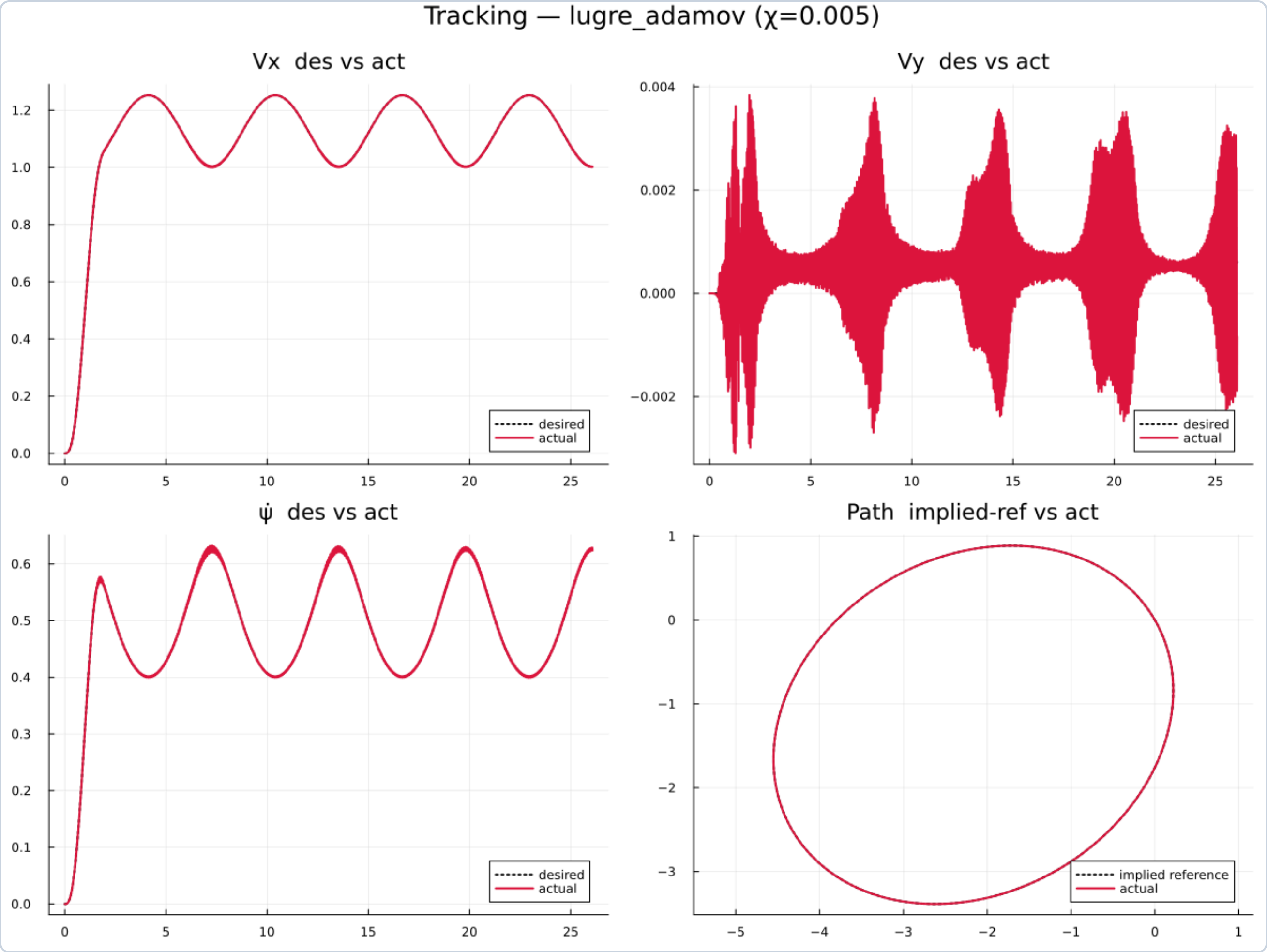
World-**position-tracked** orbit (the PosRef profile): time-varying speed *and* curvature. Heading **tangent** (forward-rich) or **fixed/crab** (strafing).

FRICTION-CIRCLE LOADING

Crab mode strafes $\Rightarrow$ heavy $F_\perp$, driving toward the hard lateral wall $V_{y,\text{crit}}$. **Tangent** mode forward-drives ($F_\parallel$).

SIGNIFICANCE

Keeps the **position controller** path alive; crab mode probes the lateral friction limit other profiles avoid.



Choosing the integrator: FBDF for the sweep

Benchmark (asmc4): which stiff solver yields PINN-grade force / moment labels

The slip–spin LuGre bristle carries a **≥500× stiffness** in its boundary layer (**tanh** smoothing); low-order methods under-resolve it — the **state** looks fine while the **labels** diverge (stiffness order-reduction).

ACCEPTANCE CRITERION

Recomputed labels $F_x, F_y, M_z, M_{\text{sat}}$ (2 kHz) must sit $\geq 10\times$ below the label-interpolation floor:

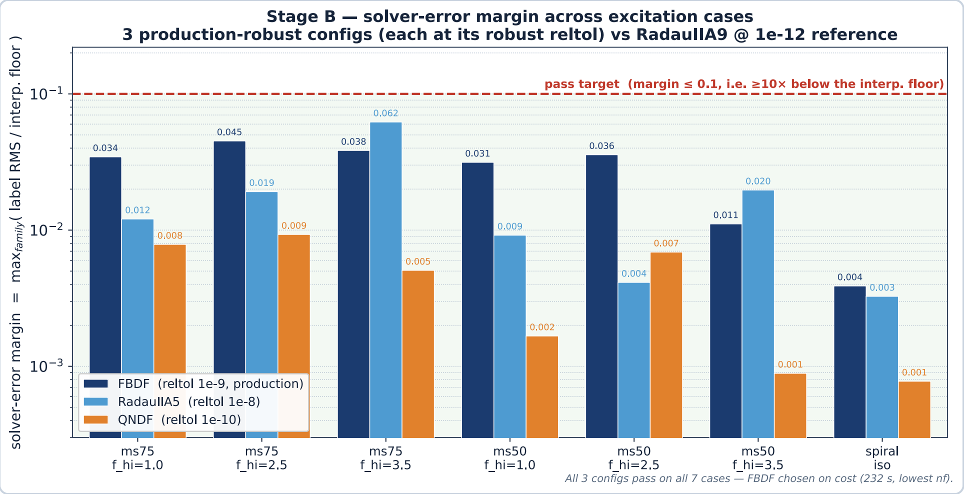
$$\text{margin} = \max_{\text{fam}} \frac{\text{err}}{\text{floor}} \leq 0.1$$

Finalists re-scored against an independent, higher-order **RadauIIA9 @ 1e–12** reference (Hairer–Wanner methodology) — removes the same-method / thin-gap objections.

CONCLUSION

FBDF @ 1e–9 is chosen for **speed** — 232 s vs 283 / 290 s, lowest n_f — while **clearing every margin** on all 7 cases. (TRBDF2, KenCarp47 reject.)

solver	reltol	margin (1e–10 ref)	margin (1e–12 ref)	verdict
RadauIIA5	1e–8	0.023	0.062	✓
QNDF	1e–10	0.062	0.0093	✓
FBDF	**1e–9**	0.062	**0.045**	**✓ chosen**
KenCarp47	—	best 1.08		✗
TRBDF2	—	best ~594		✗



Flagging trajectories unsuitable for PINN training

Two checks → pinn_training_whitelist.txt & the safe training-rate choice

1 · SPECTRAL / CHATTER SCREEN (PER WHEEL)

- **util_sat_frac** — friction / actuator budget saturation (context gate).
- **ridge_concentration** — energy on roller-ripple harmonics $h \cdot f_{\text{roll}}$ (**good**).
- **hash_fraction** — PSD energy above a LuGre-aware ceiling (**numerical hash**).
- **control_chatter_index** / **msw_force_coherence_hf** — HF switching vs force.
- **hf_slip_modulation** — HF force vs $|V_p|$ (LuGre **rescue**) · **theta_fold_tightness** — θ-periodicity.

2 · DOWNSAMPLING / ANTI-ALIASING SENSITIVITY (2 KHZ → 1 / 0.5 KHZ)

Two modes, opposite failures: **anti-aliased decimation** (no folding, but genuine HF content lost) vs **naive subsampling** $x[::k]$ (HF **folds back** in — phantom aliasing). **Flag** if band-limited reconstruction error exceeds the interpolation floor ($\sim 2.7 \times 10^{-4}$ for M_z) **or** the chatter verdict **flips** (naive aliasing manufactures 'hash'). Energy-above-Nyquist = context.

DECISION LOGIC

- **hash** (high hash, low coherence, low slip-mod) → **hard reject**.
- **chatter** (coherent ASMC switching) → **keep, flagged**.
- **clean** (ripple + LuGre only) → **keep**.
- slip-modulation **vetoes** a hash verdict — protects stick–slip.

FAILURE MODES & RATE CHOICE

chatter (keep-flag) · **hash** (reject) · **ripple / LuGre** (keep) · **saturation** (context). Check 2 sets the **safe downsample rate** per trajectory.

Solver fidelity guaranteed upstream (FBDF) — no physics-residual lane needed.

Screening the sweep — four orthogonal gates

1,776 trajectories (1,485 production sweep + 291 χ-grid) · μ=0.5, lugre–adamov, case 1

THE RECORDED "NOISE" HAS FOUR ORIGINS — TWO ARE SIGNAL

- **roller ripple** — periodic in θ, $f_{\text{roll}} \approx 1.91|\omega|$ Hz (the PINN must learn it).
- **LuGre stick–slip** — bristle oscillation, $f \approx 653|V_p|$ Hz (physical).
- **ASMC chatter** — sliding-mode switching leaking from M_{sw} (coherent, unwanted).
- **numerical hash** — broadband solver noise (incoherent, unwanted).

FOUR SCREENS → ONE WHITELIST

- **Spectral chatter** (6 metrics) + **M7 burst** — tell the four noises apart.
- **χ-identifiability** — is χ recoverable from the observable forces?
- **Sampling-rate** — can the PINN train on a coarser grid?
- **Tracking / friction-circle** — did the platform achieve the command?

Fused by precedence into **combined_reco** → the PINN-training whitelist.

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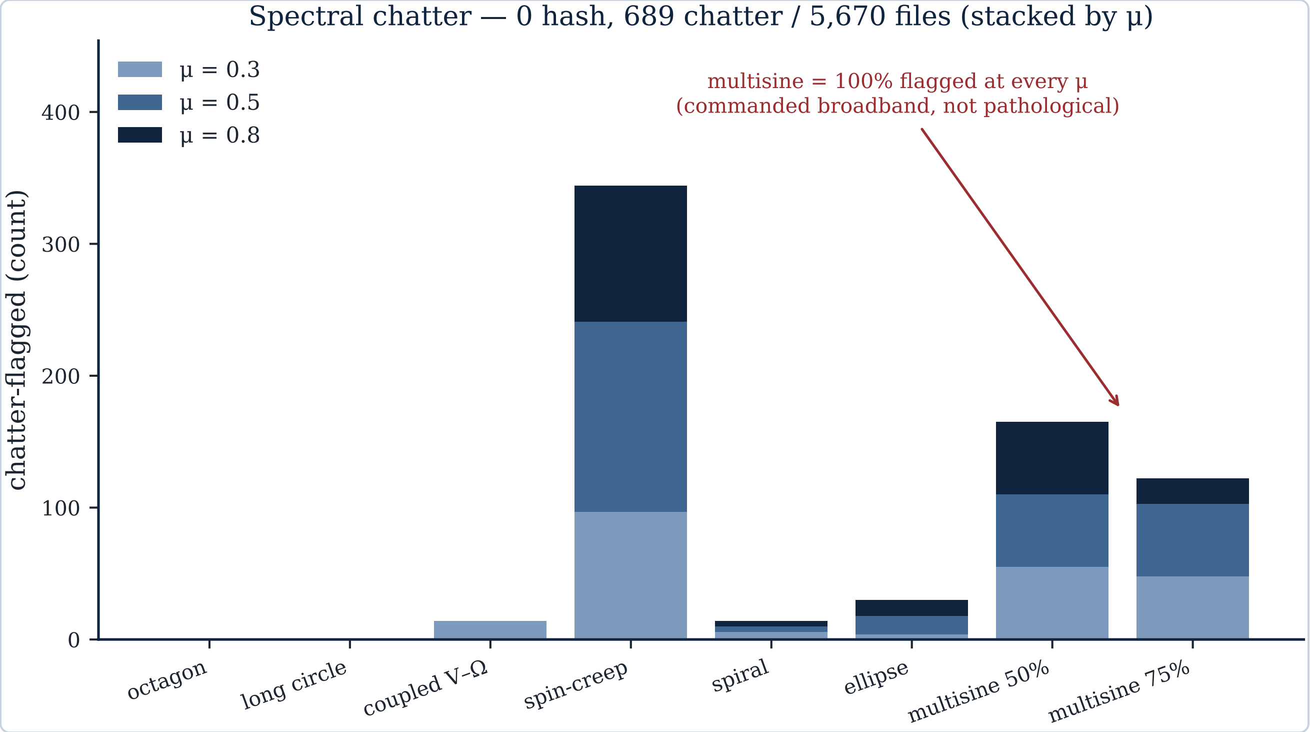
Mech. Engg.

Screen 1 — spectral chatter & the M7 burst flag

CLASSIFIER (6 SPECTRAL METRICS)

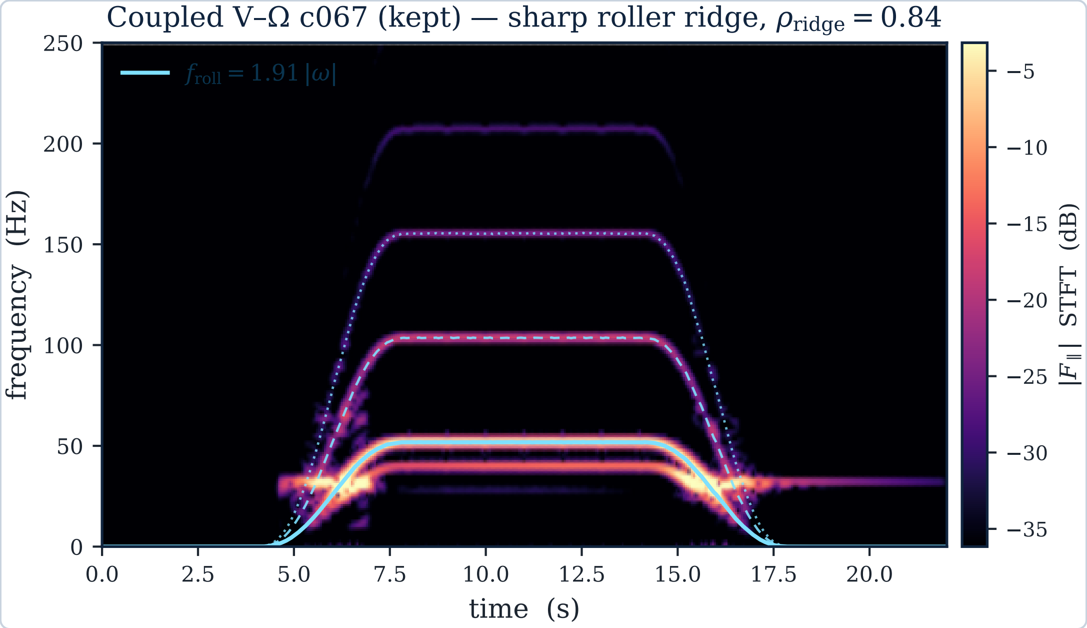
hash = energy above the ceiling, incoherent & not slip-modulated → reject; **chatter** = coherent M_{sw} leak → keep-flagged; else **clean**.

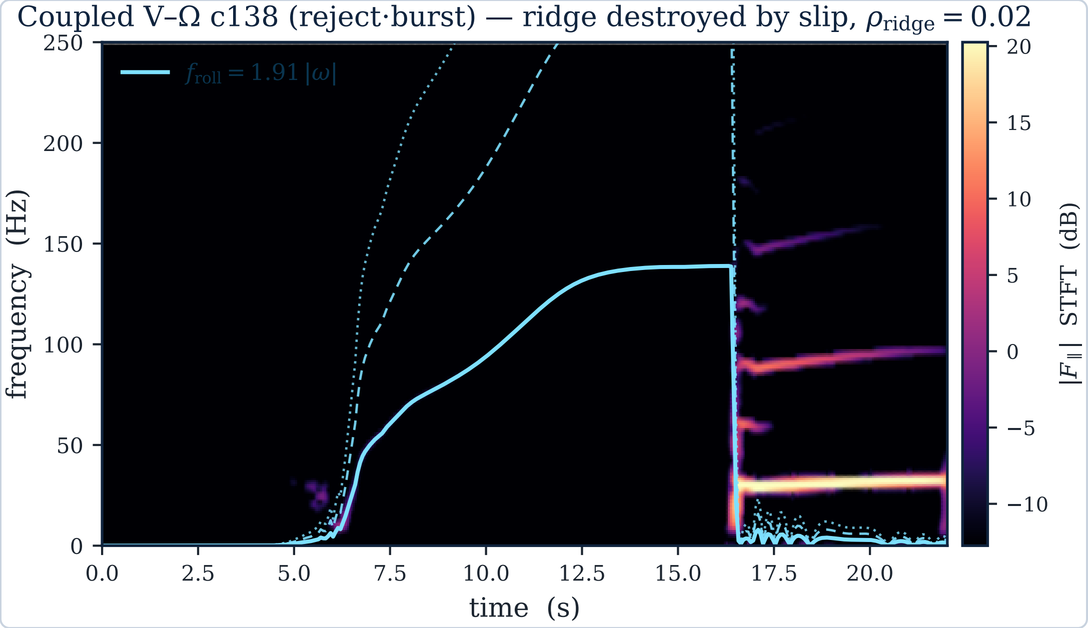
- **0 hash** across all 5,670 files (3 μ) — the stiff integration is clean everywhere. 689 chatter, 4,981 clean.
- **multisine 100%** is a **global-threshold artifact** — commanded broadband M_{sw} is coherent by design, not pathological switching (flagged at every μ).
- **spin-creep 20%**: low $|V| \Rightarrow$ thin ε boundary layer $\Rightarrow$ high K/ε limit-cycle switching (genuine).
- **M7 burst** (415): 209 give-up bursts (coupled V–Ω / octagon, degraded tracking) vs 198 spin-creep pulse edges (well-tracked) — so a burst **alone never rejects**.



The roller ridge — signal to keep vs signal destroyed

Same profile (coupled V–Ω), same channel ($F_\parallel$ STFT): the ridge-concentration metric ρ_{ridge} separates clean rolling from slip.



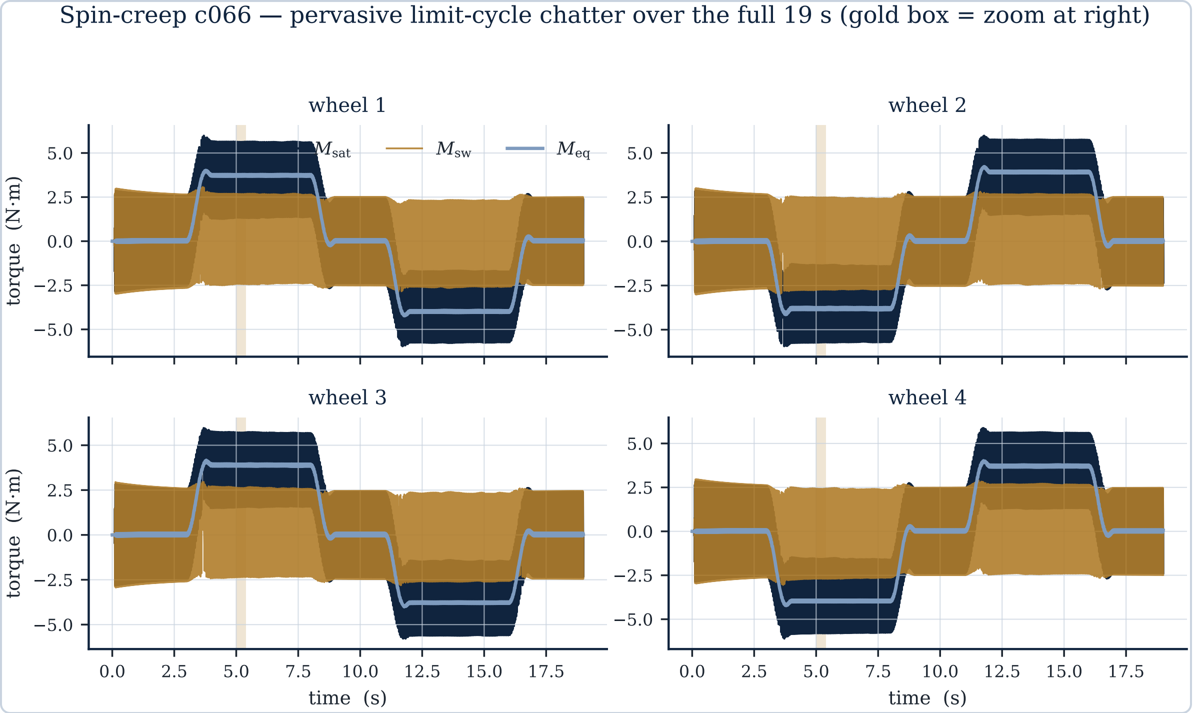


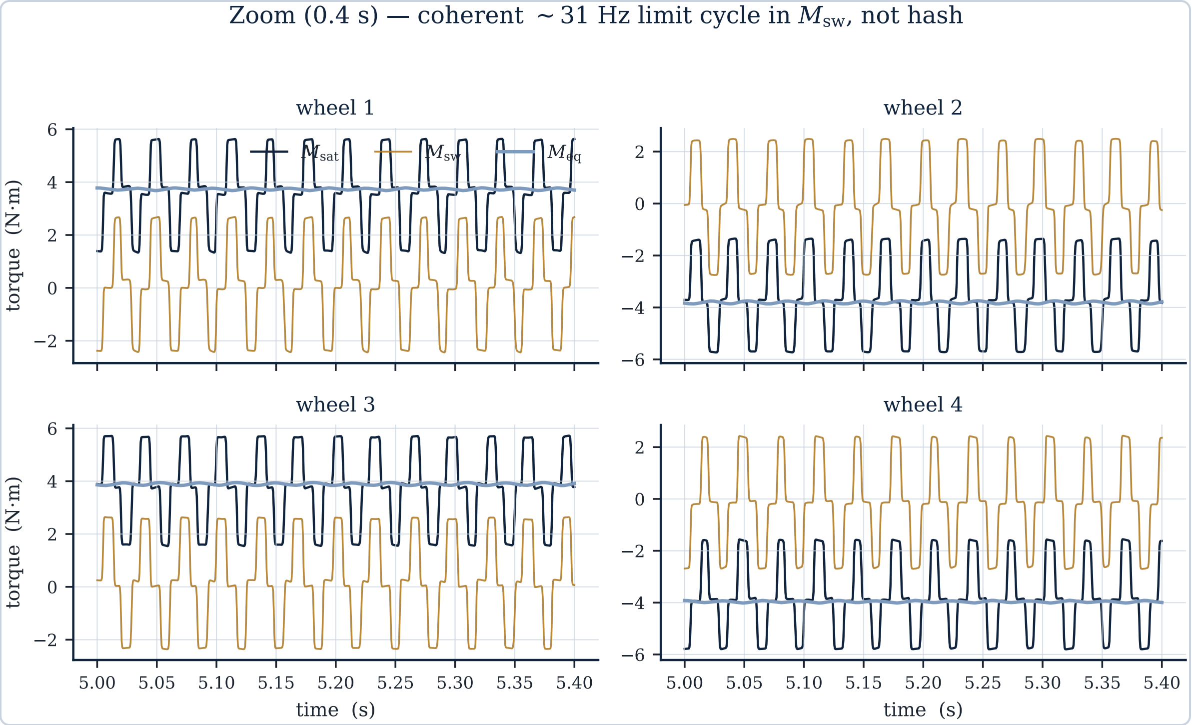
Kept (ρ_{ridge}=0.84). Energy rides exactly on $f_{\text{roll}}=1.91|\omega|$ and its harmonics as the V–Ω ray ramps — coherent, slip-locked roller ripple the PINN must learn.

Rejected (ρ_{ridge}=0.02). Over the friction circle the wheel spins (f_{roll} rising) but **slips** — no rolling ridge forms; the give-up burst ($t\approx16$ s) dumps broadband energy. The signal is gone.

ASMC chatter — coherent switching, flagged but kept

Spin-creep c066: low $|V|$ ⇒ thin ε boundary layer ⇒ high K/ε limit-cycle switching in M_{sw} — genuine, not numerical hash.

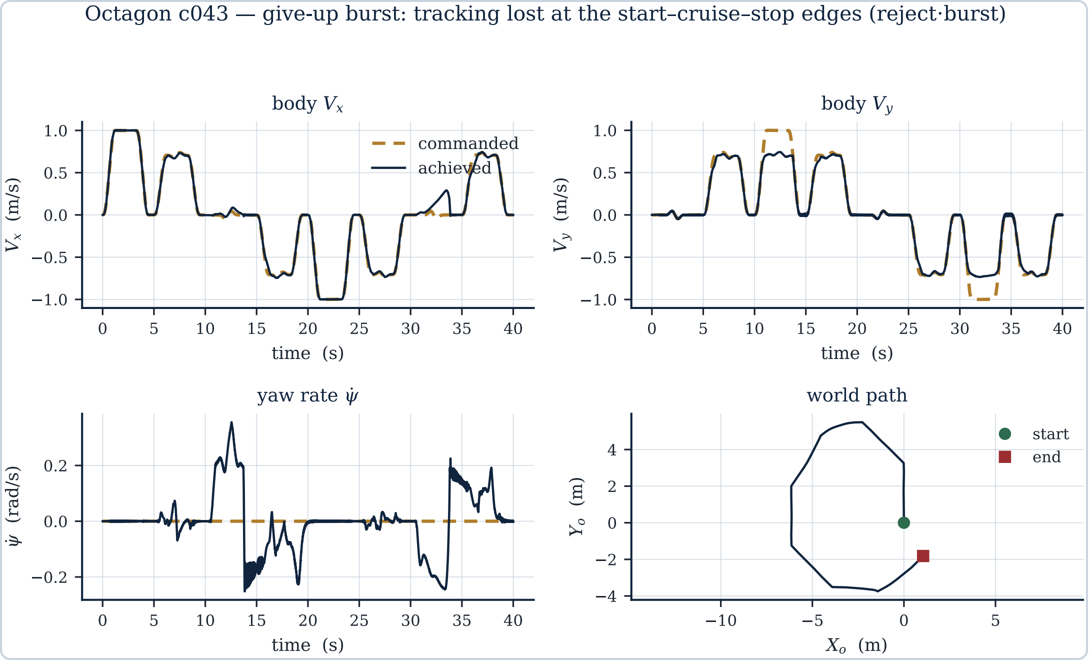




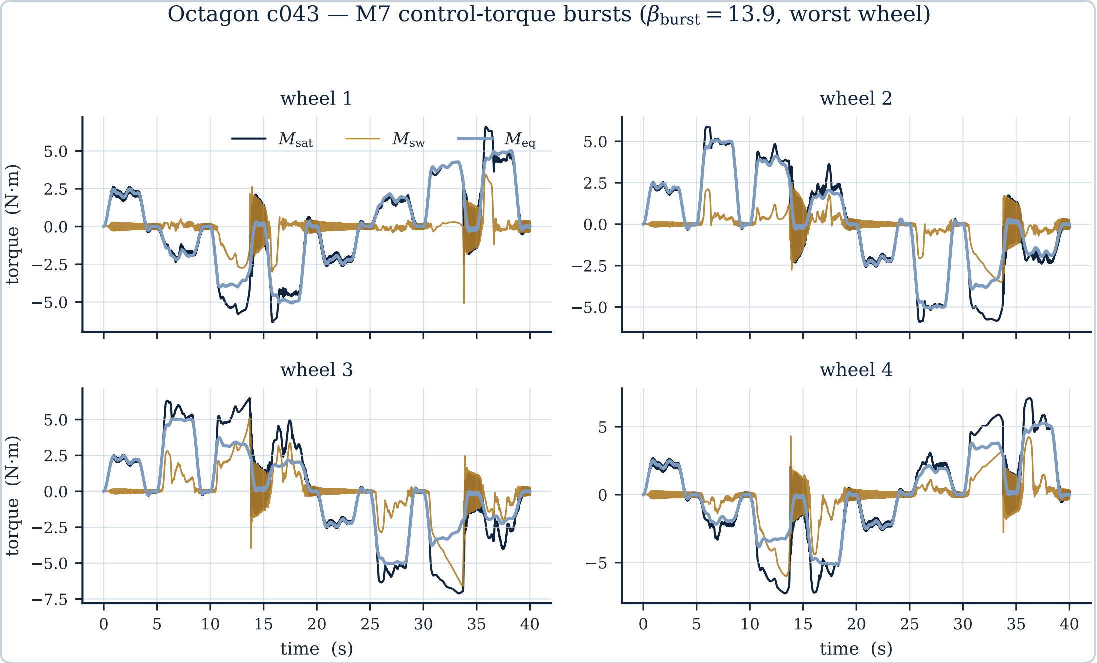
Full 19 s, all four wheels. The M_{sw} switching band (gold) persists across the whole run; M_{eq} (light) is the smooth backbone. Gold box = the window enlarged at right.

Zoom (0.4 s). A clean, periodic ~31 Hz limit cycle — coherent, not broadband → **keep-flagged**, never reject.

The M7 burst flag — reject only when tracking is also lost



Octagon c043 — uncommanded $\dot\psi$ excursions and V_y short of -1: **tracking lost**.



Localized M_{sat} bursts at the same instants ($t \approx 13, 33$ s) — $\beta_{\text{burst}} = 13.9$.

M7 BURSTINESS

15 Hz Butterworth high-pass → 0.25 s sliding-window RMS → $\beta_{\text{burst}} = \max_{\text{win}} / (p_{10,\text{win}} + \text{FLOOR})$, FLOOR $= 0.10$ N·m (1% of the 10 N·m cap), max over 4 wheels. Flag if $\beta_{\text{burst}} > \tau_{\text{burst}} = 10$. Here 13.9 ∧ lost tracking → **reject·burst**; spin-creep's 66 well-tracked pulse-edge bursts **stay**.

WHY 15 HZ, WHY p_{10}

15 Hz ≈ 5× the wheel corner $p_1/(2\pi J_w) = 0.11/(2\pi \cdot 5.87 \times 10^{-3}) \approx 3.0$ Hz: above it the wheel inertia low-passes torque (no tracking work), below the ∼29 Hz burst and ∼77 Hz roller corner. A **ratio** (multisine's flat HF $\to\!\approx 1$); p_{10} (not median) stays robust when > 50% of the run is disturbed.

VNIT Nagpur
Mech. Engg.

Screen 2 — χ is identifiable from the observable forces

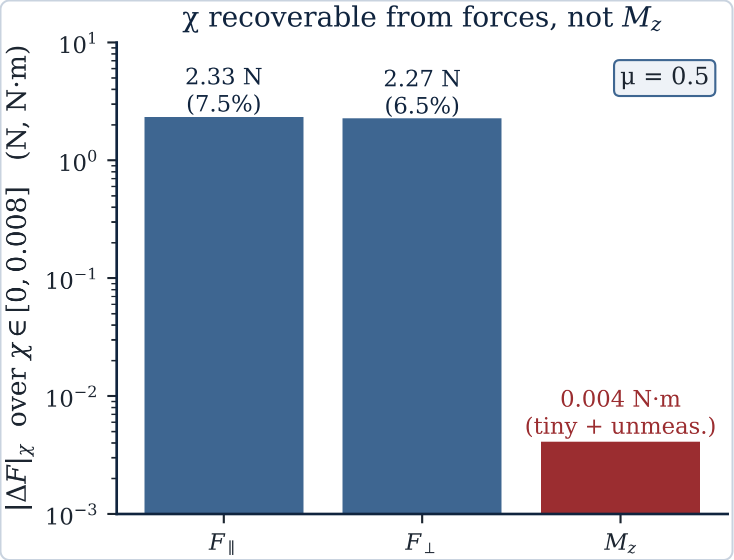
χ enters the forces only through the **spin-gated coupling** $c_t = \frac{8}{3\pi}|\omega_z|\chi$. Method: 291 matched χ-quads $\{0, 0.002, 0.005, 0.008\}$, binned by $|\omega_z|$, regressed on χ controlling for slip.

- **Recoverable from forces**: $F_\parallel, F_\perp$ swing $\approx 2.3\,\mathrm{N}$ ($\approx 7\%$ of $\sim 30\,\mathrm{N}$) at high spin — far above any force-noise floor.
- **Drop M_z**: its $|\Delta M_z| = 0.004\,\mathrm{N{\cdot}m}$ is $\sim 1000\times$ smaller *and* unmeasurable at deployment.
- The χ-effect **rises with** $|\omega_z|$ — the spin-gated mechanism, confirmed.

STRATIFIED BINNING — WHY THIS WAY

χ acts only via spin, so we **bin by contact spin** $|\omega_z|$ and, within each bin, regress each force on χ controlling for slip $(V_{px}, V_{py}, |V_p|)$ + per-wheel offset. Time-differencing fails: $\mathrm{RMS}(F_\parallel(\chi) - F_\parallel(0)) \approx 32\,\mathrm{N}$ ($> F_\parallel$) from phase **decoherence** over 20 s; binning by operating point removes it.

Why absolute swing, not R²/F: $\sim 10^6$ samples/bin make the F-stat meaninglessly "significant"; partial R^2_χ is per-channel-variance-relative, so $M_z \propto \chi^2$ inflates its R^2 (the "units trap"). Identifiability vs a real noise floor is set by the **absolute** signal — $F_\parallel$'s 2.3 N wins, M_z's 0.004 N·m loses.



Screen 3 — train at 500 Hz without losing the forces

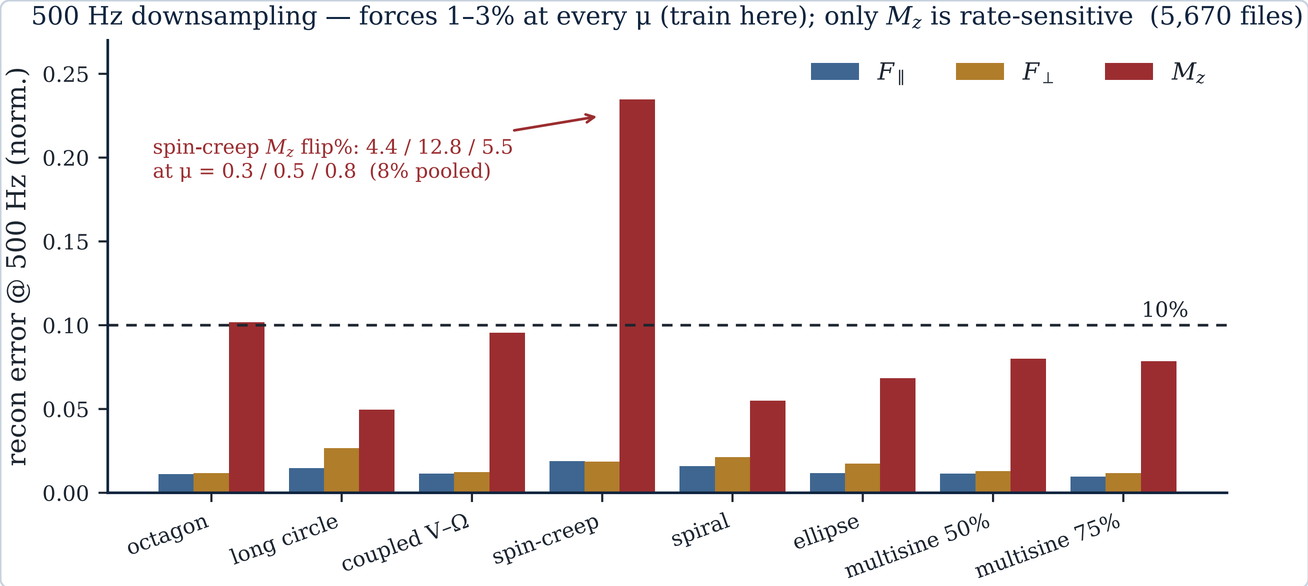
Downsample 2000 → 1000 → 500 Hz two ways; measure band-limited reconstruction error per force family.

NAIVE SUBSAMPLING VS ANTI-ALIASING

Naive = index-slice $x[::k]$: fast, but content above the new Nyquist **aliases** (folds) into the band — phantom structure. **Anti-aliased** = low-pass *then* decimate: nothing folds, but genuine HF is lost. Here they nearly agree (forces carry ~no energy >250 Hz); naive is even marginally better at 1000 Hz (the filter adds slight passband distortion) — aliasing shows up only in M_z.

- **Forces $F_\parallel, F_\perp$: 1–3%** at 500 Hz, flat from 1000→500 across every profile **and every μ → train at 500 Hz** (4× cheaper grid).
- Only **M_z is rate-sensitive** (spin-creep 0.24 at 500 Hz) — but M_z is dropped, so it's moot.
- **Screen at 2000 Hz**: the chatter verdict flips **8%** for spin-creep at 500 Hz (3-μ pooled; 12.8% at μ=0.5) — blind above the 250 Hz Nyquist.

Real forces carry almost no energy above 250 Hz (slow envelope + roller ripple ≤ 86 Hz), so the pessimistic white-input LuGre bound never bit.



Screen 4 + the whitelist — 5,345 / 5,670 kept

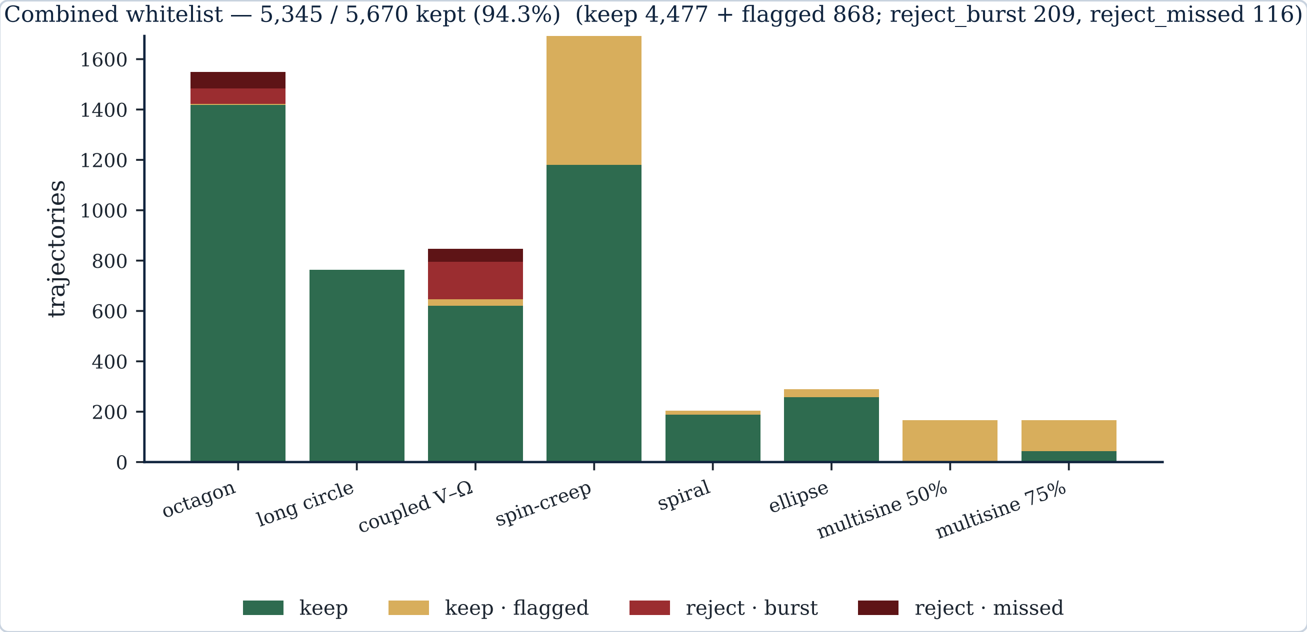
TRACKING / FRICTION-CIRCLE GATE

5,340 tracked / 5 marginal / **325 missed** (octagon start–cruise–**stop** overshoots + coupled V–Ω over the friction circle μN). **Disjoint from chatter** (a railed controller doesn't chatter).

COMBINED_RECO PRECEDENCE

① hash → **reject_hash** · ② burst ∧ lost tracking → **reject_burst** · ③ missed → **reject_missed** · else **keep** / keep_flagged. A burst must be **confirmed by lost tracking** — this keeps spin-creep's 198 pulse bursts.

keep 4,477 + flagged 868 = 5,345 / 5,670 (94.3%); reject_burst 209 + reject_missed 116. **Flat across μ** (94.8 / 94.2 / 93.8% at μ=0.3/0.5/0.8): as μ rises, reject_burst climbs (32→76→101) while reject_missed falls (67→33→16) — the gates generalise.



From plant to data generation to PINN training

Julia plant
39-D ODE: slip–spin–LuGre + ASMC

→

Excitation library
profiles.jl — 7 builders (VelRef / PosRef)

→

Seeded sweep
profile × (μ, χ, friction model) combos

↓

Solve
FBDF, reltol 1e−9, dtmax 1e−3, 2 kHz grid

→

Labels (datastore.jl)
per-wheel F, slip, controller torques

→

Arrow + JLD2
cross-language contract (filename + columns)

↓

Diagnostics
spectral / chatter + aliasing checks

→

Whitelist
pinn_training_whitelist.txt

→

■ STOP
handoff — Julia ends here

↓

PyTorch PINN (separate machine)
forward dynamics + inverse friction ID (μ, χ) · GRU encoder reconstructs hidden roller rates $\dot{\gamma}_i$ · analytical Newton–Euler integrator

χ-identifiability — spin-gated, recoverable from the observable forces

χ reaches the forces only via the spin→translation coupling $c_t = \frac{8}{3\pi}|\omega_z|\,\chi$ — linear in χ, gated by spin. Per-μ, whitelisted matched χ-quads, slip-controlled.

X-INDUCED FORCE SWING $|\Delta F|_\chi$ AT HIGH SPIN (PER FRICTION LEVEL)

μ	$F_\parallel$	$F_\perp$
0.3	1.07 N (10.0%)	1.60 N (8.4%)
0.5	1.44 N (8.1%)	2.84 N (9.2%)
0.8	1.27 N (3.3%)	1.20 N (2.6%)

- **χ is identifiable** — a ~1–3 N swing (3–10% of force RMS) at high spin sits above any realistic noise floor; **strongest at μ=0.5 / $F_\perp$**.
- **rel-swing falls as μ rises** ($F_\perp$ 8.4 → 9.2 → 2.6%): larger force at high μ, same-order χ effect → relatively harder to read.

FORCE-VARIANCE SHARE ∂R^2_χ VS CONTACT SPIN $|\omega_z|$

$\lvert\omega_z\rvert$ [rad/s]	const C	explicit $\lvert\omega_z\rvert$	χ-swing
~1.2 (low)	0.01	0.02	~1 N
~6.2	0.12	0.16	~4 N
~11.2	0.26	0.30	~6 N
~16.2 (high)	0.42	0.71	~4–12 N

COUPLED TO SPIN RECONSTRUCTION

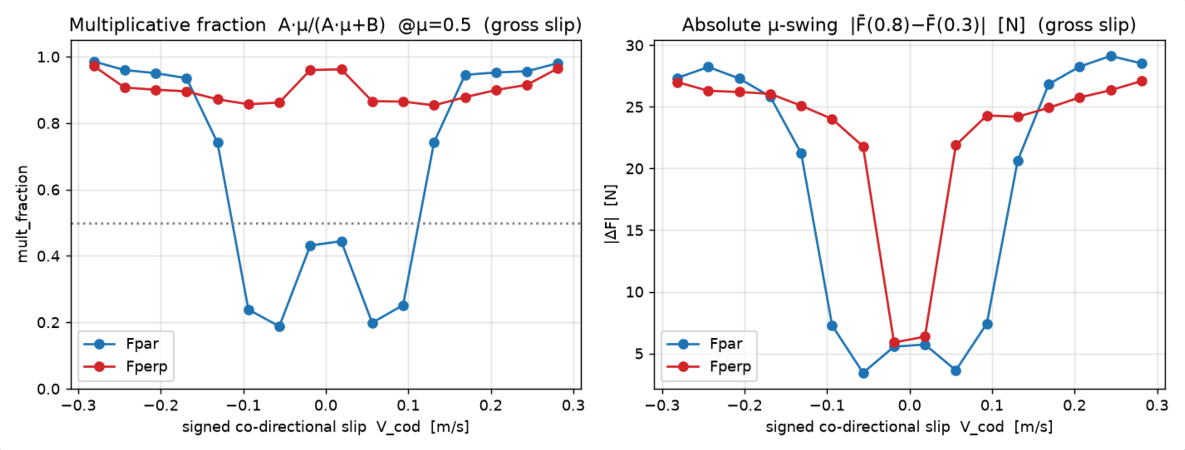
χ explains ~0–2% of force variance at low spin, **40–70% at high spin**. A constant C (blind to $|\omega_z|$) recovers only ~⅔. And $|\omega_z|$ is **~90% hidden roller rate** (§ below) → reading χ needs the encoder/observer to reconstruct contact spin.

μ-identifiability — slip-gated, multiplicative in the forces

μ enters multiplicatively, χ modulates it: $F = \mu\,(A + C\chi) + B$ (LuGre linearization; B = μ/χ-independent bristle term). Per-wheel F/N_i, binned by signed co-directional slip.

regime ($\lvert V_p \rvert$)	ch.	mult	$\lvert\Delta F\rvert$ bulk	high-slip
pre-slip <0.01	both	—	μ-blind (bristle)	
transition	$F_{\parallel}$	0.37	2.9 N	—
	$F_{\perp}$	0.80	6.2 N	—
gross ≥0.03	$F_{\parallel}$	0.24	5.5 N	~28 N
	$F_{\perp}$	0.96	6.4 N	~26 N

- **Strongly identifiable at high co-directional slip** ($\lvert\Delta F\rvert \approx 26-29$ N over μ 0.3→0.8); weak at low slip (large μ-independent intercept).
- **$F_{\parallel}/F_{\perp}$ asymmetry:** the drive axis $F_{\parallel}$ carries the structural intercept B; the free-roll $F_{\perp}$ has $B \approx 0$ → the **cleaner μ-reader**.



Within gross slip, multiplicativity is itself co-directional-slip-gated: B-dominated at small $\lvert V_{\mathrm{cod}}\rvert$ (mult ≈ 0.2) → near-pure multiplicative at large $\lvert V_{\mathrm{cod}}\rvert$ (mult ≈ 0.95–0.98, $\lvert\Delta F\rvert$ ≈ 26–29 N).

The identifiable regime is a thin tail — gate the confidence

Slip-regime coverage over 5,345 whitelisted files (1.354 B wheel-samples). Regime on total slip $|V_p|$ vs Stribeck $v_{\mathrm{str}} = 0.01$.

regime	$\lvert V_p\rvert$ [m/s]	share
pre-slip	< 0.01	**32.3%**
transition	0.01–0.03	**22.8%**
gross	≥ 0.03	**44.9%**

Per μ (gross): 46.4 / 44.7 / 43.5% — gross shrinks as friction rises (more grip). The grids span every regime — not regime-starved.

READABLE ONLY IN THE TAIL OF GROSS SLIP

~45% of data is gross slip, but only **~4.0% ($F_\parallel$) / ~4.7% ($F_\perp$)** sits at high co-directional slip $|V_{\mathrm{cod}}| \geq \mathbf{0.15}$ where μ is readable (≈60 M samples). Most gross slip is *perpendicular* free-roll — the drive axis stays near-clean.

CONSEQUENCE FOR TRAINING

The other ~95% is μ-blind by construction (affine-B bulk). So μ confidence must be **gated on** $|V_{\mathrm{cod}}|$ **/ saturation** (read μ only in the tail) or the high-$|V_{\mathrm{cod}}|$ data **tail-weighted**. χ is thinner still — it needs high co-directional slip **and** high spin → a few-percent intersection.

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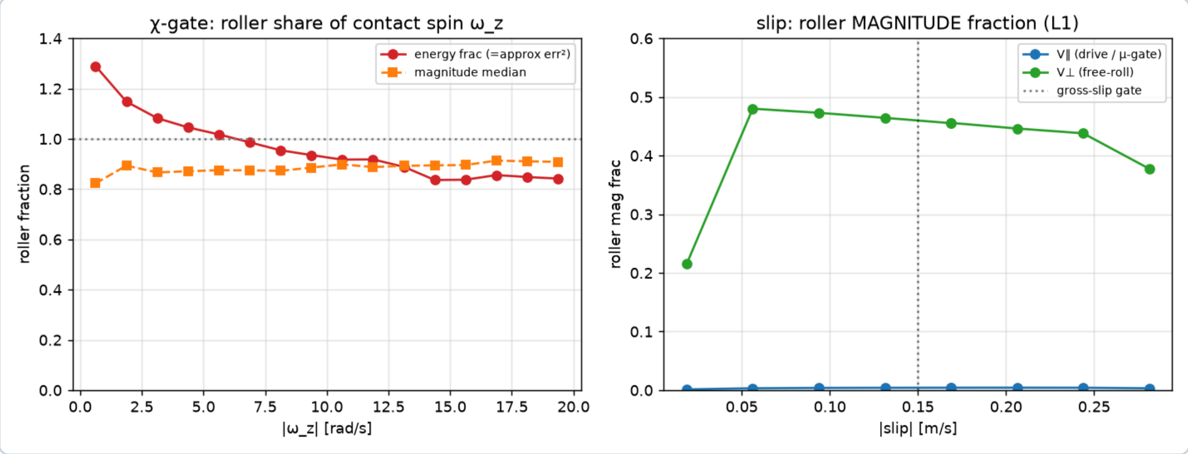
Mech. Engg.

Gate variables are hidden; the no-slip assumption is wrong



gate var	roller share
ω_z (χ-gate)	~87% energy / ~90% mag
$V_\parallel$ (μ-gate, drive)	~0.2–0.4%
$V_\perp$ (free-roll)	~24–48%

χ-gate needs the observer (ω_z ~90% hidden roller in every regime); the μ-gate rides on measurables.

FORCE ANISOTROPY $r = |F_\perp|/|F_\parallel|$

regime	median r	$r > 1$	RMS ratio
pre-slip	0.40	17.1%	0.52
transition	0.34	22.5%	0.54
gross	0.42	26.9%	0.66
ALL	**0.39**	**22.8%**	**0.62**

"FORCE ONLY ALONG THE AXLE" IS MATERIALLY WRONG

The free-roll force is typically **~40%** of the drive force, **exceeds it in ~23%** of samples, and dropping it discards **~38% of the force energy**. Modelling both channels — and the composite friction law — is justified; it is why $F_\perp$ is a usable (and the cleaner) μ-reader.

Force-law adequacy & implications for training

Can $F=\mu(A+C\chi)+B$ (with A, B, C functions of slip/spin) serve as the PINN's forward friction head? Form adequacy R^2_{form}:

channel	quad	full	gross: low→high
$F_\perp$ (free-roll)	0.76–0.80	0.71–0.78	0.86 → 0.98
$F_\parallel$ (drive)	0.26–0.35	0.11–0.16	**0.14 → 0.98**

- **The form is the right backbone, ~½ the force.** Pooled $R^2_{\text{form}} \approx 0.45-0.59$; the residual ~½ is the bristle **state** $\sigma_1 \dot{z}$ the recurrent encoder must supply.
- $F_\parallel$ **is bimodal:** ~98% algebraic in gross drive-slip, only ~14% at low drive-slip — **that** is where the SSM is mandatory. $F_\perp$ is ~75–80% algebraic everywhere.

FORWARD FRICTION HEAD (HAND-BACK TO PINN TRAINING)

$$F=\mu\big(A(\text{slip})+C(\text{slip},|\omega_z|)\,\chi\big)+B(\text{slip})$$

The analytic backbone supplies ~½ the force (most of $F_\perp$, gross-slip $F_\parallel$); an **SSM supplies the state-dependent remainder** (concentrated in low-drive-slip $F_\parallel$); and the χ channel requires **reconstructed** $|\omega_z|$.

GATED CONFIDENCE, BOTH PARAMETERS

μ ̂ read only at high co-directional slip (slip-gated); χ ̂ only at high spin **and** slip (spin-gated, observer-dependent). Both are **tail-weighted / confidence-gated** — the affine-B bulk would otherwise drown the estimate.

Visvesvaraya National Institute of Technology, Nagpur
Department of Mechanical Engineering

What the digital twin delivers

A physically-faithful, controller-in-the-loop data generator — discrete 12-roller slip–spin geometry, a C^{∞} LuGre–Adamov friction law valid in pre-slip, UUB-stable adaptive control, and a friction-circle-bounded excitation suite — producing screened, label-rich trajectories for **forward-dynamics learning** and **inverse friction force identification**.

Thank you · Questions welcome

Vishvesh Koranne
Guide: Prof. Shital S. Chiddarwar · VNIT Nagpur